中文 English

Meter Gomoku

五子棋研究与实战辅助 · 作者：Meter

下载：https://wwbbj.lanzout.com/b016l49aqj 取件密码：15

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一、软件简介

Meter Gomoku（米特五子棋）是一款在本机运行的五子棋研究与训练工具，把棋局推演、引擎评估、算杀、屏幕识图与实战连线辅助合在一起。全部计算都在你自己的电脑上完成 —— 界面与引擎以进程内 C++ 直连，识别组件走本机管道，软件不监听、也不使用任何网络端口（零 HTTP、零 WebSocket），断网也能用。

软件共有四个窗口：主窗（棋盘 + 三张功能卡）、复盘窗（只有棋盘的独立窗口，不连引擎）、连线窗（在外部对局软件里辅助实战）、识图窗（把对局截图变成局面，见第九节）。主窗与复盘窗共用同一套界面代码；连线窗与识图窗是固定尺寸的独立窗口，各自运行。

本说明按「功能」逐项展开，并配以软件界面的实际切片插图，便于对照。所有插图均为软件真机截图按界面结构切出的局部。

想做什么	去哪儿
下一盘、让 AI 陪练或复盘一局	第三节 对局设置 · 第四节 棋盘操作
看清某一手好不好、该防哪里	第五节 计算评估
算一条必胜的杀棋线	第六节 前瞻 · VCF · VCT
把一张对局截图变成可继续练习的局面	第九节 识图

想做什么	去哪儿
在对战平台 / 对手软件里让 AI 帮手	第十一节 连线功能
查开源组件与许可、找不到东西怎么办	第十三节 开源组件 · 第十二节 常见问题

表 1-1 按需求索引章节

近期主要变化

方面	变化
计算评估卡	四个键：分析计算 / 扫描分析 / 平衡点（1~2 点）/ 清除计算。四类计算同一时刻只跑一种，切到另一项会自动停掉上一项并擦掉它留在棋盘上的标记，不会再出现两套标注叠在一起
分析路线	卡片中部新增「分析路线」框：分析几点就按名次列几条，每条分两行 —— 上一行是「题目」（名次 / 分数 / 深度 d / 胜率），下一行是一整条带序号药丸的代码框。鼠标悬停某手会预览到那一手，点一下固定、再点取消；双击某手直接把这一手落到棋盘上
扫描分析	原「扫描防守」改名。扫描点集由「选点范围」决定（默认 square3_line4，形状跟棋子走，不再铺满全盘），环内直接显示该点胜率百分比；停止后最后一帧留在棋盘上，落子或「清除计算」才消失
平衡点	一次只给 1 点或 2 点（点「平衡点」右边的点数键循环切换）。标记改为细实线圆环 + 半透明紫 / 天蓝填充，环内是胜率与引擎分数，左上角序号徽标套在环外只相交三分之一；环径与棋子同大，便于贴着盘面读
分析点数	默认 1 路全强度；要看多点分析再自己加到 2~8。同机 10 秒实测：1 路到 27 层，3 路只有 25 层 —— 默认单路是为了把算力全给最可行的那一条
指导视图	整项移除（主窗、连线窗都不再有这个开关）。落点提示改由「分析计算」与「分析路线」承担
识图参数	识别参数面板收到 5 项（灰度偏差 / 黑色偏差 / 黑色占比 / 白色偏差 / 白色占比），拖动即时生效并自动保存
关于	关于页标题行右侧新增「主页」按钮，一点直达产品主页
引擎调用层	引擎命令统一走宿主队列：落子抢占、平衡与扫描互斥、分析计算长任务并行时不再互相打断，深度不会莫名其妙掉回 0；引擎按 CPU 指令集分发，纯本地计算

表 1-2 近期主要变化

二、下载、安装与运行

本软件为绿色免安装：下载后解压到任意目录（建议非中文路径，或直接用安装包装到默认位置），双击主程序即可运行，不写注册表、不装驱动。

项目	内容
名称	Meter Gomoku 连线训练器（主程序 Desktop GomokuTrainer）
下载地址	https://wwbbj.lanzout.com/b016l49aqj
取件密码	15（打开下载页后按提示输入）
适用系统	Windows 10 / 11 64 位，亦可在 Windows Server 上运行
运行方式	绿色免安装，解压即用；也提供带开始菜单快捷方式的安装包

项目	内容
授权	界面与配套代码 MIT；棋力内核 Rapfi GPL-3.0；识别组件与 OpenCV Apache-2.0
手册随附	本说明（中英双语）随压缩包一同分发

表 2-1 发行信息（下载地址与取件密码）

运行环境

项目	要求	说明
操作系统	Windows 10 1809 以上（建议 2004+ / Windows 11）	边框识别与「窗口常驻最上层」用到 Windows 10 2004 起的显示 API；更早的系统会自动降级运行，功能不受影响
界面运行时	Microsoft Edge WebView2 Runtime	Windows 10 / 11 通常已随系统预装；若缺失，安装包会提示并引导前往微软官方页面免费安装
处理器	x86-64，支持 SSE4.1 及以上	引擎内置五个指令集变体，启动时按本机 CPU 自动选择最快的一支，老机器自动降级到 SSE 版
内存	4 GB 起，建议 8 GB 以上	引擎的置换表（哈希）按可用内存自适应；内存越大、长考越稳
显卡	无要求	纯 CPU 计算，不依赖显卡，也不需要独立显卡
网络	不需要	全程本地运行，不监听任何端口，可断网使用

表 2-2 运行环境与依赖

目录说明

路径	内容
Desktop GomokuTrainer.exe	主程序：界面宿主、引擎调度、识别调度
Vision/	识图组件 GomokuVision.exe 及其静态链接库（含 OpenCV）
rapfi-native/	棋力内核 Rapfi 的五个指令集变体（avx512vnni / avxvnni / avx512 / avx2 / sse）、神经网络权重、开局与配置，以及 GPL 许可文本与来源说明
calc/	界面资源：加密后的页面、样式与脚本（防止误改，正常运行不需要动它）
软件说明 (Software Description).pdf	本说明（中文版在前、英文版在后，可点目录与书签）

表 2-3 目录结构与各文件的作用

三、对局设置

主窗右侧的「对局设置」卡片分四个可折叠板块：对局、引擎、棋盘显示、局面。点板块标题即可折叠 / 展开。

（一）对局板块

这一板块决定「谁和谁下、下多久、按什么规则」。



图 3-1 对局设置 · 对局板块（模式 / AI 执子 / 思考时间 / 后台思考 / 动态时间 / 规则）

模式

模式	AI 参与	典型用途
对弈	是（按 AI 执子设置自动应答）	实战、复盘、让 AI 陪练；这是日常使用最多的一种
摆盘	否	自由摆子构图，研究布局或试做变化图；各类计算提示停摆
残局	否	摆一个局面专门研究，配合棋盘下方的摆盘键与「确定」，详见第四节

表 3-1 三种模式的差别

AI 执子

两个开关可任意组合，四种组合各有明确含义：

AI 执黑	AI 执白	谁先走	说明
开	关	AI 先走（天元）	你执白。AI 落第一手天元，之后按你的落子应答
关	开	你先走	你执黑。AI 等你落子后应答
开	开	双方都是 AI	AI 自打：你只旁观，棋盘点击无效；点棋盘正中的键可暂停 / 继续
关	关	你	双人手动对弈：没有 AI，黑白双方都由你自己下

表 3-2 AI 执子四种组合

思考时间

项目	取值	说明
可调范围	0.2 ~ 300 秒，步进 0.1 秒	直接输入数字即可，默认 2 秒；短到闪电出手，长到深度长考
棋力关系	时间越长越强	AI 候选热力与扫描分析的色档也随思考时间自动调节
与连线窗同步	同一个口径	连线窗里的「思考时间」就是这一项，两边改任意一侧都会同步
与后台思考的关系	互不占用	后台思考只用对手思考的空档预热，不从这个时间里扣
实际消耗比例	约 92%	引擎在用到设定时间的 92% 时收手，留出余量把最后一层迭代算完；再往上压往往只是让最后一层被中途砍断，并不会更强

表 3-3 思考时间的取值范围与影响

后台思考

卡上的「后台思考」开关：打开时对手思考期间引擎就在算你的应对（改用引擎原生 pondering，与官方整合包同一套节奏），你一落子它已就绪；关闭时引擎落子后立刻停下，不再空转高占用 CPU（默认关闭）。

动态时间

「动态时间」按局面难易分配每一手的思考时间，而不是每手固定秒数。判断依据全部来自上一手引擎自己的输出（评估分、是否已见杀、首选着法是否稳定），不额外搜索、不增加延迟。省下的秒数存进「时间池」，难局面从池里借；池的上下限是 −1×N ～ +2×N —— 借到 −N 就再也借不出，自动退化成「每手就按你设的 N 走」，不会滚雪球。

局面（上一手的结果）	系数	说明
已见杀（+M35 / −M34）	0.40	执行或防守已经明确，再算也是浪费
大优（评估 ⩾ 2500）	0.70	胜负已分
明显优势（⩾ 900）	0.85	方向明确，只需微调
大劣（⩽ −1500）	0.80	能守的选择很有限
均势且首选着法还在变	1.15	真的难 —— 从时间池里借
连续两手同一着法	0.85	结论已收敛

表 3-4 动态时间：系数与触发条件

实测（思考时间设 2 秒）：均势的几手实际用 2.2~2.6 秒，见杀局面 0.1~0.2 秒就落子。关掉这个开关即回到「每手严格 N 秒」，时间池同时清零。

规则

规则	禁手	换色	要点
无禁手	无	无	连成五子（含长连）即胜，最自由
无禁手 · 长连不赢	无	无	标准规则：长连不算赢，必须正好五子
连珠（有禁手）	黑有	无	黑方三三 / 四四 / 长连为禁手，白方无禁手；禁手点会画红叉
一手交换	无	有	黑下第 1 手后，白方可选择是否与黑交换颜色
山口规则 Yamaguchi	有	有	黑摆 3 子（黑1 天元、白2 限 3×3、黑3 限 5×5）并声明第 5 手的打点数；对方选色后白4 任意位置，黑按声明数叫点（对称点不可重复），白挑一个落第 5 手
塔拉山口规则 Taraguchi	有	有	黑1 天元、白2 限 3×3、黑3 限 5×5、白4 限 7×7；第 5 手可直接下（9×9）或十打叫 10 点由对方挑，随后对方选色
塔拉尼科夫规则 Tarannikov	有	有	黑1 天元、白2 限 3×3、黑3 限 5×5、白4 限 7×7、黑5 限 9×9；每手之后对方均可换色，无打点

表 3-5 七种规则的禁手、换色与要点

前瞻推演与算杀（VCF / VCT）都会跟随当前所选规则；交换类规则在开局时会弹出小窗，让你选先手或后手。切换规则时不会清空棋盘 —— 保留当前布局，按新规则继续，终局判定与禁手标记会重新计算。

（二）引擎板块



图 3-2 对局设置 · 引擎板块（线程 / 哈希 / 引擎变体 / 权重 / 选点范围 / 分析点数 / 最大深度 / 节点）

线程 引擎的搜索线程数。档位按本机真实拓扑给出，旁边标注「物理核 / 逻辑线程 · 推荐 · 上限」；默认停在推荐值（混合架构只算性能核），散热与调度跟得上时可自行拉到上限（逻辑核减二）

哈希 引擎置换表（TT）的大小。正在干活的那条引擎车道独占所选档位的 100%，其余车道降到地板值，因此总占用始终等于你所选的档位

引擎变体 「自动（推荐）」按实测吞吐从各指令集构建里挑最快的二进制

权重 「NNUE（精确）」棋力优先；「classical」在快棋里更快

选点范围 引擎的 CAUTION_FACTOR，六档：0 最激进（5×5 范围）… 5 最谨慎（全盘），默认 3（7×7+4，官方推荐档）

分析点数 1 ~ 8，默认 1（单路全强度）；与计算评估卡的分析点数共用同一个值，两端双向同步

最大深度 / 最大节点 可选硬上限；保持默认即不限制搜索

Active / Idle / Usage 主车道与辅助车道的哈希分配：主车道吃满所选档位，辅助车道为地板值

表 3-6 引擎板块各字段

（三）棋盘显示板块

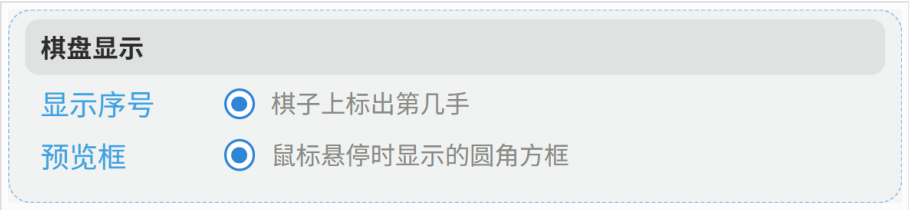


图 3-3 对局设置 · 棋盘显示板块（显示序号 / 预览框）

「显示序号」在每颗棋子上标出它是第几手，复盘时最有用；「预览框」是鼠标悬停时显示的圆角方框，默认开启，关掉后落子更「素」。

（四）局面板块

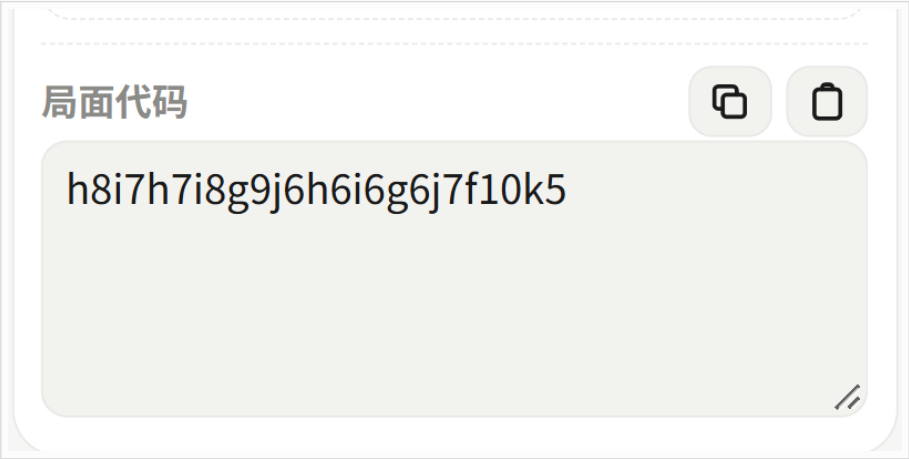


图 3-4 对局设置 · 局面板块（局面代码的复制 / 粘贴 / 载入）

局面代码用小写列字母 + 行号逐手拼接（与 gomocalc.com 同格式），例如 **h8i9g7** 表示依次落子于 H8、I9、G7。可「复制 / 粘贴 / 载入」，用于导入导出、分享或存档一个局面。

（五）AI 执子小窗



图 3-5 「AI 执子」小窗（棋盘下沿功能键可随时弹出）

四、棋盘操作与残局摆盘

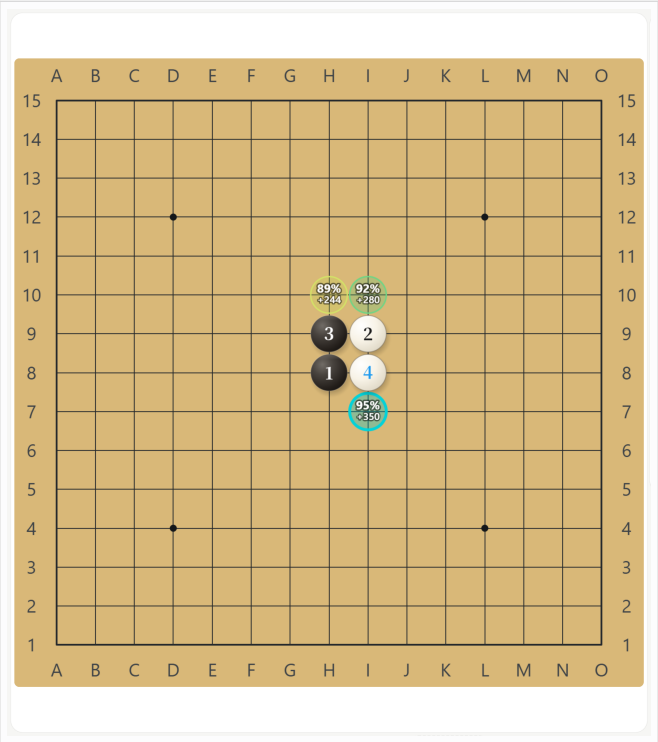


图 4-1 主窗棋盘：上沿坐标轴，下沿功能键行，棋盘正中一颗「继续 / 暂停」键

操作	方式	说明
落子	直接点击棋盘交点	当前行棋方由落子数的奇偶推出，无需手动切换
上一步 / 下一步	棋盘下沿 ‹ › 键	逐手回退与前进，用于复盘
重新开始	棋盘下沿「重新开始」	会先把当前这一局自动存进历史，再清盘
保存局面	棋盘下沿「保存局面」	把当前盘面手动存入历史
旋转	棋盘下沿「旋转」	棋子整体顺时针旋转 90°
翻转	棋盘下沿「翻转」	弹出小窗可选左右 / 上下 / 两种对角翻转
平移	棋盘下沿「平移」	局面整体平移一格；若有子会出盘则整体不动并给出提示
让 AI 立刻走一手 / 暂停	棋盘正中的键	空闲时点它 = AI 立刻为当前行棋方算一手并落下；AI 思考中点头 = 暂停
AI 执子 / 设置	棋盘下沿最右两键	分别弹出「AI 执子」与「设置」小窗

表 4-1 棋盘操作速查

旋转 / 翻转 / 平移三者只变换棋子布局，棋盘线与坐标轴不动。

残局摆盘

把模式切成「残局」后，棋盘下方会出现摆盘键，用来把一个局面精确摆出来再研究：

摆盘键	用法
顺序摆盘	黑白轮流落子，按落子顺序摆放
任意摆盘	点开右侧圆角小框选「黑子」或「白子」，之后可连续摆该色，数量与顺序都不限；选中的色键会变色
确定	点最左的「确定」把局面定下来：键变浅绿并显示「已确定」。此后「重新开始」回到该局面，存进历史时它也是首帧；再点一次即可解除重摆

表 4-2 残局摆盘三个键

残局模式下 AI 全程不参与，交换手类规则自动变灰（残局没有开局交换可言）。

五、计算评估

这一卡是软件的「棋力工作台」，自上而下分四层：最上面四个键 —— 分析计算 / 扫描分析 / 平衡点（带 1~2 点切换）/ 清除计算；接着是分析路线框，分析几点就列几条完整路线；中间是前瞻 · VCF · VCT 与逐手代码链；最下面是分析点数。四类计算同一时刻只跑一种 —— 切到另一项时，软件会自动停掉上一项并擦掉它留在棋盘上的标记，不会出现两套标注叠在一起。



图 5-1 计算评估卡：分析计算 / 扫描分析 / 平衡点（1~2 点）/ 清除计算 · 分析路线 · 前瞻 · VCF · VCT · 分析点数

分析计算 在主车道持续深算当前局面，棋盘上铺出引擎当前榜单（最佳点 + 候选，最多 8 个），右侧「分析路线」同步按名次列出路线，随搜索加深自然收敛；再点一次即停止。摆盘、残局与双人手动对弈下同样可用 —— 只要有一个「你」的执子方存在就能铺榜单

扫描分析 走引擎原生的 YXSEARCHDEFEND（与官方整合包同一算法）：对「选点范围」覆盖到的每一个点深算，环内报出的百分比就是「对手下在这里」的胜率，越高越该防。点集形状由选点范围决定（默认 square3_line4，跟着棋子走，不是死板的整盘方块）；按钮随后变成「停止分析」，停止后最后一帧留在棋盘上，落子或「清除计算」才消失

平衡点 走官方平衡搜索（引擎原生 YXBALANCETWO 成对搜索）：给出黑白双方都没有明显优势的候选点，并把平衡引擎自己那一对的评估分一并报出 —— |评估| ⩽ 50 才是官方平衡窗口内的「平」（刻度 ScalingFactor=200 ⇒ 胜率 = sigmoid(评估/200)）。标记是细实线圆环 + 半透明填充，环内上半是胜率、下半是分数，左上角序号徽标套在环外

点数 平衡点一次给几个，点「平衡点」右侧的点数键循环 1 点 → 2 点：1 点是单点平衡（五手两打里的「打」），2 点是成对的两个平衡点；空盘固定 3 手

分析路线 卡片中部的框：分析点数设成 N，这里就按名次列出 N 条。每条分两行 —— 上一行是「题目」（名次 / 分数 / 深度 d / 胜率），下一行是一整条带序号药丸的代码框（与引擎仪表盘的路线代码同款）。鼠标悬停某一手会预览到那一手，点一下固定、再点取消；双击某一手直接把这条路线实落到那一手

前瞻 · VCF · VCT 见下一节。三个键在本卡的中段，旁边是逐手代码链与「确定」，下方一行是进攻方指示框

清除计算 一键擦掉本卡任意项目留在棋盘上的标记（分析计算的名次与分数、扫描分析的环与百分比、平衡点的环、前瞻 / VCF / VCT 的序号与虚线），只清标记，不动局面本身

表 5-1 计算评估的各项功能

项目	取值与含义
分析点数	1 ~ 8，默认 1；与「对局设置 · 分析点数」是同一个设置，两端双向同步。默认单路是为了把算力全给最可行的那一条 —— 同机 10 秒实测：1 路到 27 层，3 路只有 25 层
平衡点数	1 点 / 2 点，点一下循环；空盘固定 3 手。点数越多，平衡搜索的预算越分散
扫描分析着色	按该点胜率分档上色：危险度越高环越深越实、越低越浅；环内直接写百分比数字
分析路线	一行「题目」+ 一行代码框；题目给出名次、分数、深度（d 值）与胜率
单模式互斥	分析计算 / 扫描分析 / 平衡点 / 前瞻四类只能同时开一个，切换时自动收尾并清标记

表 5-2 计算评估的关键参数

这四类计算共用同一台引擎，但走的是不同通道（分析计算走长任务主车道、扫描分析走 YXSEARCHDEFEND、平衡点走 YXBALANCETWO）。因此切模式时软件会先把上一个作业停干净再起新的，避免两条通道互相打断。

六、前瞻 · VCF · VCT

功能	说明
前瞻	从当前局面逐手推演，一路走到盘面真的五子连珠才结束 —— 不凭引擎的胜负判断提前定性，最坏情况是铺满棋盘判和。首手会加强全局计算找突破口，并给出预计耗时区间；推演子是半透明的、数字表示手数；每一步都做必应校验（能连五就连五、对手成五必堵、活三必应）
VCF	连续冲四必胜搜索：只考虑连续冲四的强制序列
VCT	冲四 + 活三连续威胁搜索：在冲四之外把活三也纳入威胁序列
进攻方	黑 / 白为指示框（不可点），自动判定 = 轮走方；算杀题面按惯例取子多的一方
代码链	左右步进键可逐手查看；悬浮代码预览到那一手；点击选中、「确定」把这一段落到棋盘
十打	叫 10 个候选点，交给对方挑选

表 6-1 前瞻与算杀六项

VCF 与 VCT 都跑本机算杀器（多线程并行 + 迭代加深），命中即给出必胜线并在棋盘上标注杀路；无解会弹窗明示「所选进攻方不存在该强制杀」。选了 VCF / VCT 后，你在棋盘上落子时它会边走边回探，一旦能接手就切进杀棋序列。

七、引擎仪表盘与评估曲线



图 7-1 引擎仪表盘与评估曲线卡

深度 当前搜索到达的深度（层）

速度 每秒搜索的节点数，反映算力利用是否正常

节点 本次搜索已展开的节点总数

用时 本次（分析计算模式下跨轮累加）的累计用时

评估 当前局面对行棋方的评分，正数占优、负数落后

最佳 当前搜索给出的最佳一手

评估曲线 按黑 / 白两条记录整局走势；纵轴随分数自动进位（最大约 ±1200），横轴可滚动查看

状态行 轮到你 / AI 思考中 / 暂停 / 终局 / 背诵等提示（最多两行）

表 7-1 仪表盘与曲线字段

任意计算期间（AI 落子、分析计算、扫描分析、平衡点、前瞻、VCF、VCT）这些数值都会边算边跳，是判断「引擎有没有在干活、干得顺不顺」最直接的地方。

八、历史与复盘

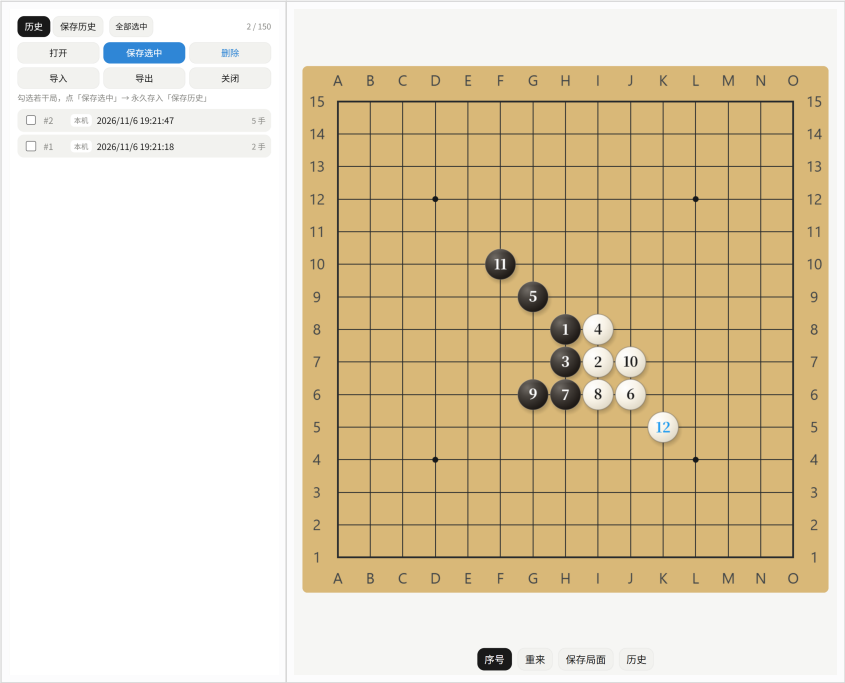


图 8-1 历史抽屉（左）与复盘窗（右）：抽屉管存档，复盘窗只有棋盘

按键 / 操作	作用
历史 / 保存历史	抽屉内两个页面：前者是自动存档，后者是你手动保存的对局
打开	把选中的一局载入棋盘（VC / 残局记录以整盘为首帧，不必从第一手重放）
保存选中	把选中的记录搬进「保存历史」页
删除 / 全部选中	删除选中项；「全部选中」一键勾满当前页
导入 / 导出	支持 txt 格式的导入与导出
Ctrl / Shift 多选	像文件管理器一样连续或离散多选
右键菜单	对单条记录：打开复盘 / 重命名 / 导出这一局 / 存入保存历史 / 删除
复盘窗	顶栏「复盘」打开一个只有棋盘的独立窗口，不连 AI、不参与其他功能；可用「背诵复盘 / 回顾复盘」逐手走

表 8-1 历史抽屉与复盘窗操作

九、识图（Gomoku Vision）

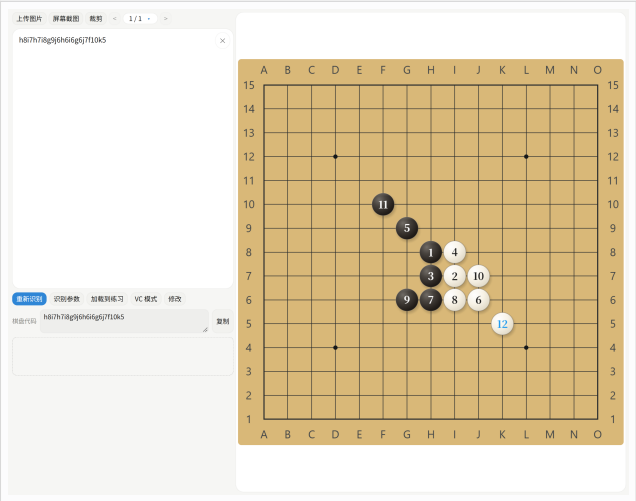


图 9-1 识图面板：上传 / 截图、框选、识别、人工修改与图片抽屉

上传 / 截图 导入一张对局截图；也可以直接抓取屏幕区域

框选 选框会自动贴住棋盘，并向外多留约棋盘 1/30 的余量，确保最外侧的棋子完整进框；识别前也可先手动框选图片区域

识别 点「识别」开始；结果会同步成局面代码，可复制，也可直接载入练习

修改 人工微调：交换黑白子（点一黑一白即互换）、删除棋子、补充黑子 / 白子；Ctrl+Z 撤销

识别参数 面板上的「识别参数」把 5 项阈值（灰度偏差 / 黑色偏差 / 黑色占比 / 白色偏差 / 白色占比）直接摊在面板上，拖动即时生效；任一项偏离默认值即走严格判定，结果完全按你设的参数来；提示行固定高度，拖滑动条时整列不会抖

图片抽屉 每次上传的图片自动进抽屉（超过 150 张清最老），支持多选、删除与复制

表 9-1 识图的使用步骤

识图对「看着能认却认不出来」这类情况做了专门加强：整盘读空时先按本帧峰值放松阈值重判，仍为空就换一套局部对比读子（只看交点中心与外圈对角小块的亮度差，深色盘、浅色盘都不再漏）；找盘先后尝试原图、去 JPEG 块效应、CLAHE 均衡、1.5 倍放大四档；网格与线条没对上时还会请旧识别器（投影 + 部分格点，对旋转容忍得多）复核，谁更合理用谁；整行或整列 15 格同色会判为网格错位一格并自动平移重读。

对比项	普通识图	VC（算杀题）模式
黑白子数	必须相等	可以不等
补子 / 删子	受棋局约束	补黑 / 补白不限数量，删除任意删
载入后	按局面继续	自动切成残局模式，并预设进攻方（子多的一方）
算杀	不自动开	也不自动开 —— 点 VCF 或 VCT 由你自己决定

表 9-2 VC 模式与普通识图的差别

十、规则、语言与外观

项目	说明
规则	七种，见第三节「对局设置 · 规则」的对照表
语言	顶栏「EN / 中」一键切换全界面语言，包含识图截图框的按键与历史里的日期格式
颜色	调色窗管理棋盘色、页面底色、深色 / 浅色主题以及预制色方案
布局	右栏卡片可用标题栏拖动换位、点小横杠折叠、点关闭键隐藏；布局自动记忆，下次启动照旧
棋子材质	设置 → 字体 → 棋子 三档：仿真（立体渐变 + 落盘微阴影，默认）、玉石（层状浅杂色 + 玻璃光泽）、扁平（纯色快绘）
棋盘清晰度	棋盘按 2 倍超采样渲染，棋子、手数、网格线与坐标在高分屏上不发虚；阴影用预烘焙贴图，重绘约 0.4~0.6 ms/帧

表 10-1 规则、语言与外观

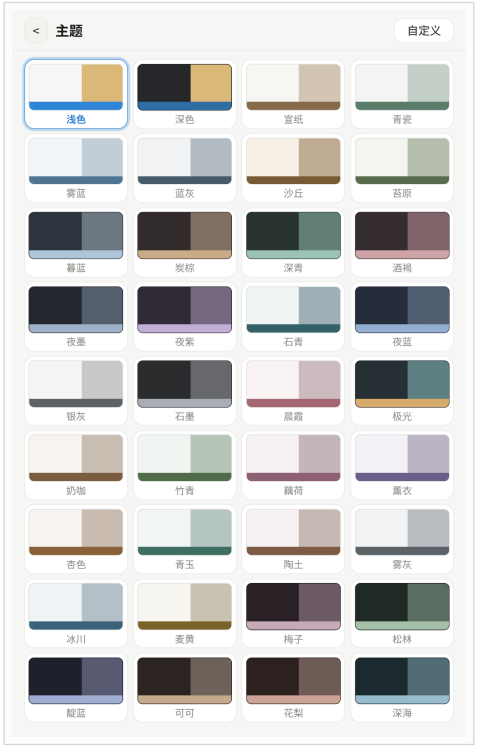


图 10-1 调色窗：棋盘 / 页面底色、深浅主题与预制色

十一、连线功能（在外部对局软件中实战辅助）

连线功能是整个软件的核心：它把「看棋盘」和「算棋」都放进一个固定尺寸的小窗里，通过屏幕识别读到你正在对局的那盘棋，再让引擎帮你想。它不接触对方软件的任何内部接口 —— 只做屏幕取像与模拟点击两件事。



图 11-1 连线窗全貌（左：整窗；右：控制面板放大）—— 窗内自带上沿栏、控制面板、状态区与右侧局面小棋盘

（一）框选与识别

按键	作用
框选棋盘	在屏幕上拖出一个矩形框住目标棋盘；框完成后窗内开始识别
识别参数	把上面这块按键区就地换成 5 项识别阈值（灰度偏差 / 黑色偏差 / 黑色占比 / 白色偏差 / 白色占比），拖动即时生效；「应用」保存并收起，「取消」回到上次已应用的值
真实点击	用真实鼠标点击落子（默认关闭）。只认真实鼠标输入的程序（如 QQ 五子棋）才需要打开；打开期间会占用你的光标
自动落子	打开后识别到对方落子即自动思考，并用模拟点击替你落子（详见下节）
×1	落子次数：每个落点连点几下，1 ~ 3 循环切换（标题提示当前档位）

表 11-1 框选与识别相关按键



图 11-2 连线窗上沿：框选棋盘 / 识别参数 / 真实点击 / 自动落子 / ×1

识别全程在后台静默进行，分两步：首次框选用软件自带的识别把整盘定型（同时给出 15 条横线、15 条竖线的精确位置），锁定之后改由识别引擎的同步读子接管 —— 交接时把窗口指纹一起交过去，所以全盘只搜一次、下一帧就能接上。此后按逐格变化工作：每个交点用 6×6 采样签名缓存，画面静止时

0 格重读、几乎不花时间；落一子只重读落子那一格。因此框色不会来回闪，也不会持续占用 CPU。自动裁框时会向棋盘外多留 0.8 格的余量（方形棋盘的最外侧一枚子最容易被切掉，这个余量保证它完整进框且不会把相邻 UI 误收进来）。

识别对图片的适应性也做了三层兜底（仅在五项阈值全为默认时才启用；任一项非默认即走严格判定）：整体读不到棋子时，先用「本帧峰值」放松阈值重判一次；仍为空就换一套局部对比读子（只看交点中心与外圈对角小块之间的亮度差，深色棋盘、浅色棋盘一视同仁）；网格与线条没对上时（liveness 过低）还会请旧识别器（投影 + 部分格点，对旋转容忍得多）再算一遍，谁更合理用谁。整行或整列出现 15 格同色这类不可能结果时，会自动平移一格重读。

识别参数



图 11-3 「识别参数」：点一下就把上面的按键区换成 5 项阈值，边拖边生效

参数	作用
灰度偏差(0-255)	判定「这里有棋子」的灰度门限。调大更容易判成棋子，调小更严格
黑色偏差 / 黑色占比	黑子的灰度门限与格内占比。黑子识别过多 → 调小；漏识 → 调大
白色偏差 / 白色占比	同上，针对白子
参数严格生效	任意一项偏离默认值即切换到严格判定：结果按你设的参数出，不再被松阈值 / 局部对比 / 网格复核等兜底链静默改写；只有全部保持默认时才启用兜底
通用 / 存档 / 读档	「通用」一键载入默认值（50 / 96 / 33 / 96 / 33）；「存档 / 读档」把当前这 5 项按主题名存起来（最多 50 个），读档抽屉里可加载 / 全选 / 删除 / 撤销
应用 / 取消	「应用」保存并收起参数区；「取消」把 5 项还原成打开时那一份，宿主侧的有效阈值也一并还原

表 11-2 识别参数的五项阈值与四个按键

（二）控制面板

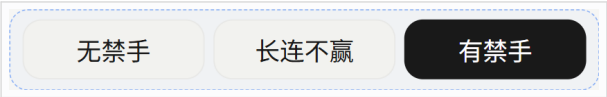


图 11-4 规则三键：无禁手 / 长连不赢 / 有禁手

控件	说明
规则三键	无禁手 / 长连不赢 / 有禁手，直接切换连线窗使用的终局与禁手判定
框选棋盘 / 识别参数 / 真实点击 / 自动落子 / ×1	见上一节
AI 执黑 / AI 执白	互斥的二选一（也可以都不选，由你自己下）；选择会无条件同步到主窗对局设置里的执子
思考时间	0.2 ~ 300 秒，与主窗同一个口径（两边始终一致）
交换平衡点	给出开局平衡点（见下节）
交换评估	与交换平衡点相互独立的另一项功能，用于评估当前交换局面的优劣，判断是否值得交换
取消	中止在途的交换平衡 / 交换评估计算，并清掉已经画出的平衡标记，回到正常的连线状态

表 11-3 连线窗控制面板按键



图 11-5 AI 执黑 / AI 执白与思考时间（互斥档，选中会同步到主窗）

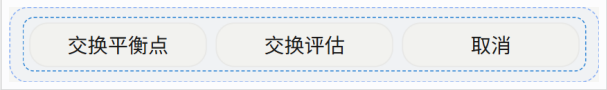


图 11-6 交换平衡点 / 交换评估 / 取消（「取消」只在计算或标记存在时可用）

（三）状态区与局面小棋盘



图 11-7 状态区：深度 / 速度 / 节点 / 评估

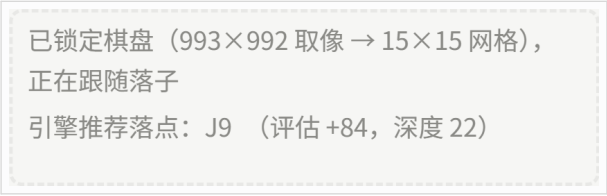


图 11-8 状态文字区（虚线大框，留足三行以上余量，中英文换行长也不折叠）

区域	内容
状态行	深度 / 速度 / 节点 / 评估，任意计算期间边算边跳
状态文字	说明当前正在做什么（识别中 / 思考中 / 已标出平衡点等）；打开「识别参数」时这里改成常显 5 项参数的调节提示，可以上下滚动翻看
局面小棋盘	实时镜像识别结果，旁边显示手数；语义与助手的局面小窗一致 —— 识别到什么就显示什么，没识别到就保持上一手好盘面，不无故清空

表 11-4 连线窗状态区与局面小棋盘

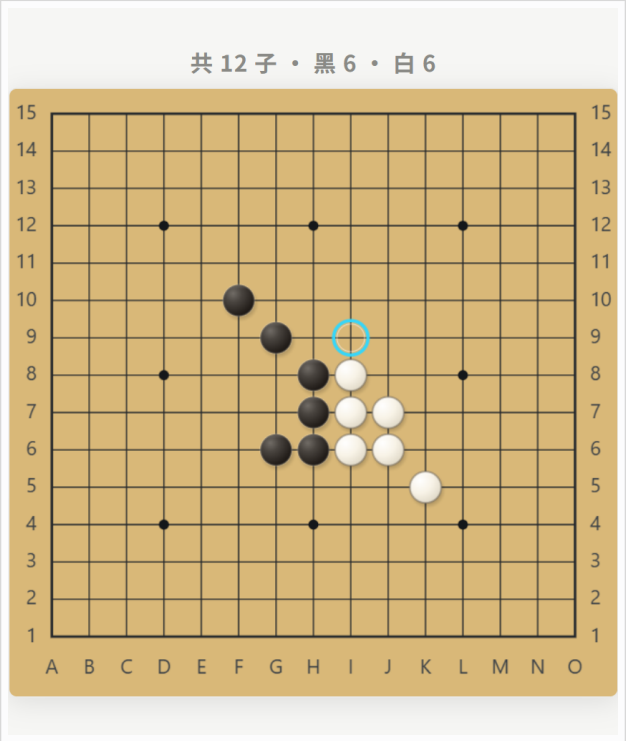


图 11-9 右侧局面小棋盘（实时镜像识别结果）与手数

（四）分工：谁负责什么

连线窗负责识别与 AI 的全部工作（评估、自动落子）；主窗的大棋盘只同步显示识别到的局面与执子设置，本身不参与评估、不落子。这样两侧永远不会抢同一份算力 —— 打开连线时主窗自动切到 1.16 : 1 的布局，关闭连线后主窗恢复你自己原来的模式与设置。

（五）自动落子

级数	方式	触发条件
一级	PostMessage 直达目标窗口	默认首选：最轻、不干扰物理鼠标
二级	SendInput 模拟输入	一级回帧判定失败后自动升级
三级	SetCursorPos + mouse_event 真实点击	前两级都无效时的兜底；成功后会被「钉住」复用
裁决	回帧确认	每一级都用下一帧判断落子是否真的生效，未生效才升级
重试	整轮重来（最多 3 轮）	三级全失效不会就此死等：自动重开一整轮，三轮都失败才停机提示
收尾	开局即自动结束	一局结束（五子连珠）自动关闭自动落子，下一局需重新打开

表 11-5 自动落子的三级点击管线与重试

打开自动落子时若你执黑且棋盘是空的，它会先替你下天元。物理鼠标全程不受干扰 —— 三级管线里只有最后一级会短暂借用光标，且用完即还原。

（六）交换平衡点

这是给「交换开局」用的：软件替你算出一组对双方都尽量公平的平衡开局点，并画在识别框内的对应交点上。空盘那一手走的是官方开局定式 —— 先请引擎按当前规则给出三手开局（无禁手用 Swap2 公式库、有禁手用 RIF 公式库；引擎不可用时退到软件内置的同源公式表：无禁手 9 组、禁手 26 组，且只收平衡定式、必胜定式一条不收），再用棋盘的八种对称变换（恒等、主对角、左右、上下、180°、顺 / 逆时针 90°、副对角）展开去重，实际可用型数远多于表内直接列出的那几组。三点摆出来后，还会让平衡引擎给这一对打分并一起报出来 —— 报出的才是可核对的 |评估|（实测两组公式分别落在 1~17 与 5~19，都在官方 balanceWindow(50) 以内）。

盘面手数	给出点数	颜色	说明
0 手（空盘）	3 点	两黑一白	交换开局的第一阶段：黑 1 天元固定，白 2 与黑 3 由官方定式给出
3 子（共 5 手）	2 点	一黑一白	第二次交换：一次成对平衡搜索同时给出这个黑点与白点（成对才是官方口径的「平」，单点搜索在近乎空盘上会偏到 −350~−470）
超过 5 子	不调用	—	已过交换窗口，功能不可用并给出提示

表 11-6 交换平衡点在不同手数下的行为

联动行为	说明
与 AI 执子互斥	点「交换平衡点」的同时会自动取消已选的 AI 执黑 / AI 执白（两者会相互干扰）
与自动落子联动	先点平衡点、后开自动落子：评估完成时若自动落子已开，就按序自动把这几点落下；若是评估完才打开自动落子，则打开的那一刻按序补摆
「取消」键	算到一半不想等了：立刻中止在途的平衡计算、清掉平衡标记与排队中的摆点，回到正常的连线状态（不再接着把标记画完）
标记生命周期	标记留在识别框内不消失；平衡评估引擎随即释放、不再占用算力；你把这几个子下完，标记自动消失

表 11-7 交换平衡点的联动逻辑

（七）盘中标记

标记	外观	含义
推荐落点	天蓝圆圈	引擎给出的推荐点
交换平衡点 · 黑	紫圈 + 序号	两黑一白里的黑方落点，序号 1 / 2 / 3 表示落子顺序
交换平衡点 · 白	蓝圈 + 序号	两黑一白里的白方落点
误读防护	全部排除出截图	所有标记都直接画在识别框内的交点上，并被排除出屏幕捕获 —— 绝不会被自己的识别器误读成棋子

表 11-8 棋盘上的各类标记

（八）性能：算力怎么分

连线时识别与引擎要同时跑，软件把两套负载拆到不同的物理核上，因此长时间对战也不容易卡顿或掉棋力。

子系统	占用	说明
引擎搜索	约 2/3 的物理核	按物理核心而非逻辑线程切分，避免超线程争抢；具体线程数显示在引擎板块的「本机线程数 / 上限」里
屏幕识别	其余约 1/3 的物理核	识别只在有变化时才读子：画面静止时几乎不占用 CPU，落一子只重读那一格
引擎哈希	等于所选档位	正在干活的主车道独占 100% 置换表，其余车道降到地板值，所以总占用恒等于你在对局设置里选的档位；档位上限按本机内存自动折算
长算稳定性	自动收敛	长时间连算时，标记与评估曲线只保留最近的固定条数，不会越算越吃内存

表 11-9 引擎与识别的算力分配

（九）窗口常驻最上层

连线窗本身固定在桌面最上层：它以「顶层窗口」创建，并在窗口位置变化时把自己压回最上层，显示与设备状态变化时也会重新压一次 —— 因此它不会被对局软件的窗口盖住。你自己点最小化时软件不干预，只在你把它叫回来时重新置顶。

收尾	说明
关闭连线窗	主窗自动恢复你自己原来的模式与设置，对弈 / 计算等功能照常使用
退出软件	一并关闭所有引擎与相关进程，不在后台残留

表 11-10 连线窗的关闭与收尾

十二、常见问题

现象	原因	处理
识别不到棋盘，或识别不稳定	框选没有把四角与边框完整包住	重新「框选棋盘」，可适当多框一点边缘；棋子较满时软件会结合棋子与边框共同判断
棋盘边缘的子识别不到	自动裁框余量不足，最外侧一枚被切掉	自动裁框本就向外多留 0.8 格（方形棋盘最外侧最易受影响的正是这一枚）；如仍出现，请把框选范围连棋盘外框一起框进去
自动落子开着却一直不落子	某一轮三级点击都没生效	三级（PostMessage → SendInput → 真实点击）全失效后会自动重开一整轮，最多 3 轮；三轮都失败才停机提示。若仍不动，先点「框选棋盘」重新锁定
棋力时高时低	引擎与识别争抢同一批 CPU / 内存，偶尔被饿住	软件按物理核把算力 2/3 : 1/3 拆给引擎与识别，哈希表按本机内存自适应折算；把思考时间调长也立竿见影
偶见「引擎响应超时」	极少数发生在引擎抢占 / 换档的瞬间	软件会在约 1.5 秒后自动重评并静默重试，不影响正常落子
一直提示正在识别	空心选窗或首次预热中	关闭空心选窗后重新框选；首次打开会做识别预热，稍候即可
点了「交换平衡点」没有反应	当前手数不在交换窗口内	交换平衡点只在空盘（两黑一白 3 点）或已有 3 子（一黑一白 2 点）时可用；超过五手会明确提示「不在交换窗口」
是否需要联网	—	不需要：软件与引擎全部在本机运行，不监听任何网络端口
退出后是否残留进程	—	退出时会一并关闭所有引擎与相关进程，不在后台残留

表 12-1 常见问题与处理

十三、开源组件与许可

本软件由多个组件构成，各组件依其原始许可证发布。下表列出随软件分发的全部第三方 / 开源组件、其许可证与官方链接，便于核对与获取源码。

组件	用途	许可证	官方链接
Meter Gomoku 界面与配套代码	界面、宿主程序与配套脚本（作者：Meter）	MIT	随本说明附带许可声明

组件	用途	许可证	官方链接
Rapfi	棋力内核（五子棋 / 连珠引擎），随本软件分发五个指令集变体与权重	GNU GPL-3.0	github.com/dhbloo/rapfi
OpenCV	棋盘与棋子识别，静态链入 GomokuVision.exe，不额外分发 DLL	Apache-2.0	opencv.org
Microsoft Edge WebView2 Runtime	桌面界面运行时，由 Windows / Edge 提供，本软件不随包分发	专有（微软）	microsoft-edge/webview2
思源宋体 / 思源黑体（Source Han Serif / Sans）	本说明的排版字体，随文档嵌入	SIL OFL 1.1	adobe-fonts
Inno Setup	制作 Windows 安装包的工具（仅用于打包，不随软件分发）	Inno Setup License（允许商用免费）	jrsoftware.org/isinfo.php

表 13-1 随软件分发的第三方 / 开源组件一览

许可证	允许 / 要求 / 链接
MIT	允许免费获得、使用、复制、修改、合并、发布、分发、再许可与销售副本；要求在副本中保留原版权声明与许可声明；软件按「现状」提供，不附任何明示或默示担保
GNU GPL-3.0	允许免费获取、使用、修改与再分发；要求分发时保留许可证并提供源码获取方式（Rapfi 源码见 github.com/dhbloo/rapfi，许可证全文见 gnu.org/licenses/gpl-3.0）
Apache-2.0	允许商用与再分发；要求保留版权与许可声明、标注修改；许可证全文见 apache.org/licenses/LICENSE-2.0
SIL OFL 1.1	允许免费用于个人与商业用途，包括嵌入文档再分发；要求不得单独出售字体文件本身，不得以原字体名销售改造版本；全文见 openfontlicense.org
专有（WebView2）	按微软自身条款使用；本安装包不对其再分发，仅在缺失时引导前往官方页面下载

表 13-2 各许可证的要点与全文链接

合规说明

（1）Rapfi —— 本软件以编译后的独立进程方式调用 Rapfi，双方只通过标准输入输出与消息通信，不与 Rapfi 合并为同一作品；随本软件分发 Rapfi 时已一并保留其 GPL-3.0 许可文本与来源说明（位于 rapfi-native/ 目录）。

（2）OpenCV —— 以静态链接方式并入 GomokuVision.exe，未额外分发 OpenCV 的动态库；其 Apache-2.0 许可要求保留版权与许可声明，已在识图组件处保留。

（3）字体 —— 本说明与软件界面不使用任何非商用授权字体（例如微软雅黑、黑体等 Windows 系统字体），全部改用同源开源的思源宋体 / 思源黑体。

（4）开发期依赖 —— 开发与构建期使用的脚本及其依赖（NumPy、Pillow、PyMuPDF、reportlab 等）只在本机运行，不随软件分发，因此不在上表之列。

（5）免责声明 —— 本软件按「现状」提供，作者不对软件的适用性、稳定性或任何直接 / 间接损失承担责任。请在遵守对弈平台规则的前提下合理使用辅助功能。各组件名称与商标归其各自权利人所有。

作者：Meter 界面与配套代码以 MIT 许可发布，棋力内核 Rapfi 以 GPL-3.0 发布，识别组件与 OpenCV 以 Apache-2.0 发布，文档排版使用思源宋体 / 思源黑体（SIL OFL 1.1 开源授权）。下载：https://wwbbj.lanzout.com/b016l49aqj 取件密码：15

中文 English

Meter Gomoku

Gomoku Study and Live-Play Assistant • Author: Meter

Download: https://wwbbj.lanzout.com/b016l49aqj Password: 15

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1. Introduction

Meter Gomoku is a Gomoku (Five in a Row) study and training tool that runs on your own computer. It combines position analysis, engine evaluation, mate search, screen recognition and live-play assistance. Every calculation happens locally: the UI and the engine talk to each other in-process through C++, the recognition component runs through a local pipe, and the app listens on no network port at all (zero HTTP, zero WebSocket), so it works fully offline.

There are four windows: the main window (board plus three feature cards), the Review window (a standalone board-only window that never talks to the engine), the Link window (assistance inside another playing app) and the Image window (turns a screenshot of a game into a position; see section 9). The main and Review windows share the same UI code; the Link and Image windows are fixed-size windows that run on their own.

This manual describes the software feature by feature, illustrated with real UI slices cut along the structure of the interface, so that the text and the screen always line up.

What you want to do	Where to look
Play a game, practise against the AI, review a game	Section 3 Game Setup / Section 4 Board Operations
See how good a move is, or what to defend	Section 5 Calculation and Evaluation
Search a forced winning line	Section 6 Lookahead, VCF and VCT
Turn a screenshot of a game into a position you can study	Section 9 Screen Recognition
Get help while playing inside another app	Section 11 Link Mode
Check open-source components, or fix a problem	Section 13 Licences / Section 12 FAQ

Table 1-1 Chapter index by need

Recent highlights

Area	Change
Analysis card	Four keys: Analyze / Scan / Balance (1-2 points) / Clear. Only one of the four runs at a time - switching to another stops the previous job and wipes the marks it left on the board, so two sets of annotations never end up stacked on top of each other
Routes	New Routes box in the middle of the card: N analysis points give N ranked routes, each in two lines - a "header" (rank / score / depth d / win rate) and a full pill-style code line. Hovering a move previews it, a single click pins it, and a double-click plays that move onto the board
Scan	Formerly "Scan defence". The scanned point set now follows the candidate range (square3_line4 by default, so the shape follows the stones instead of tiling the whole board) and each ring prints its own win rate as a percentage; stopping keeps the last frame until you move or press Clear
Balance	Gives 1 or 2 points only (the key next to Balance cycles them). The mark is a thin solid ring with a translucent violet / sky-blue fill, the win rate and engine score sit inside the ring, and the rank badge sits outside it, overlapping by one third; the ring is the size of a stone
Analysis points	Defaults to one full-strength PV; raise it to 2-8 for multi-point work. Measured on this machine at 10 s: one PV reaches 27 ply, three reach 25 - the single default gives all the budget to the most playable line
Coach view	Removed entirely (neither the main nor the link window has the switch any more). Move hints are now covered by Analyze and Routes
Recognition params	The recognition-parameter panel is down to 5 sliders (gray offset / black offset / black ratio / white offset / white ratio); dragging applies immediately and saves itself
About	The About sheet gained a Home button on the title row, opening the product page
Engine call layer	Engine commands go through one host queue: move pre-emption, balance and scan exclusivity and the long-running live analysis no longer interrupt each other, and the depth no longer drops back to 0; variants are dispatched per CPU instruction set, all locally

Table 1-2 Recent highlights

2. Download, Install and Run

The app is portable: unzip it anywhere (an ASCII path is safest, or just use the installer and keep the default location) and double-click the main program. It writes no registry keys and installs no drivers.

Item	Details
Name	Meter Gomoku Link Trainer (main program: Desktop GomokuTrainer)
Download	https://wwbbj.lanzout.com/b016l49aqj
Password	15 (enter it on the download page when asked)

Item	Details
Platform	Windows 10 / 11 64-bit; also runs on Windows Server
Install	Portable - unzip and run. An installer is also provided that adds a Start-menu shortcut
Licensing	Interface and companion code MIT; playing core Rapfi GPL-3.0; recognition component and OpenCV Apache-2.0
Manual	This document (bilingual) ships inside the package

Table 2-1 Release information (download link and password)

Runtime requirements

Item	Requirement	Notes
Operating system	Windows 10 1809 or later (2004+ / Windows 11 recommended)	Border recognition and the always-on-top window use display APIs from Windows 10 2004; on older systems the app downgrades automatically without losing features
UI runtime	Microsoft Edge WebView2 Runtime	Normally already present on Windows 10 / 11; if it is missing the installer points you to the free Microsoft download page
Processor	x86-64 with SSE4.1 or later	The engine ships in five instruction-set variants and picks the fastest one for your CPU at start-up, falling back to the SSE build on older machines
Memory	4 GB minimum, 8 GB or more recommended	The engine transposition table scales with available memory; more memory means steadier long searches
Graphics	None	Pure CPU calculation - no GPU and no discrete card required
Network	Not required	Everything runs locally; no port is opened, so it works offline

Table 2-2 Runtime environment and dependencies

Folders

Path	Contents
Desktop GomokuTrainer.exe	Main program: UI host, engine scheduling, recognition scheduling
Vision/	Recognition component GomokuVision.exe and its statically linked library (including OpenCV)
rapfi-native/	Playing core Rapfi in five instruction-set variants (avx512vnni / avxvnni / avx512 / avx2 / sse), its neural-network weights, opening and configuration files, plus the GPL licence text and source notice
calc/	UI resources: encrypted pages, styles and scripts (kept encrypted so they cannot be edited by accident)
软件说明 (Software Description).pdf	This manual (Chinese first, English after; clickable contents and bookmarks)

Table 2-3 Folder structure and what each file does

3. Game Setup

The "Game Setup" card on the right of the main window has four collapsible sections: Game, Engine, Board Display and Position. Click a section title to fold or unfold it.

(a) The Game section

This section decides who plays whom, for how long, and under which rules.

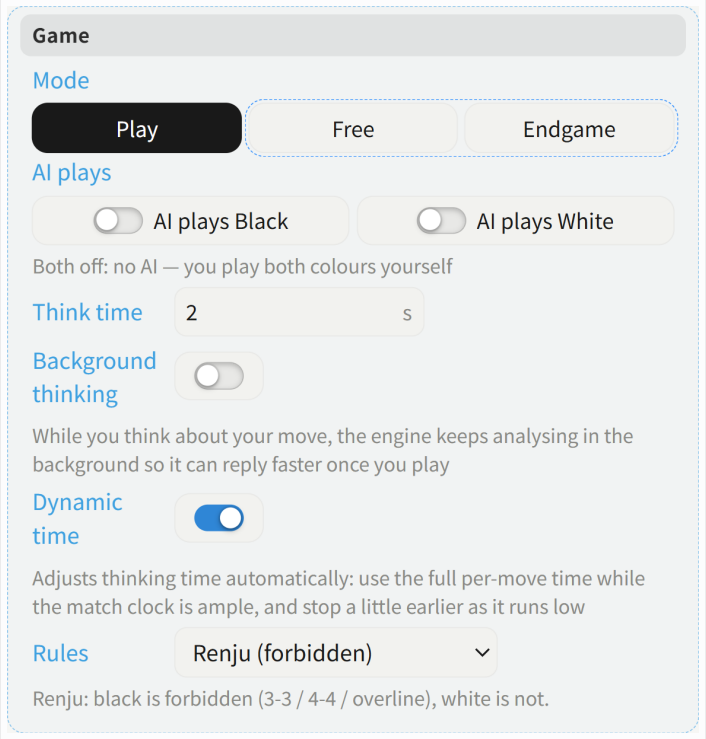


Figure 3-1 Game Setup - the Game section (mode / AI side / thinking time / background thinking / dynamic time / rules)

Mode

Mode	AI plays?	Typical use
Match	Yes (answers automatically per the AI-side setting)	Real play, review, practising against the AI - the everyday mode
Place	No	Free stone placement to study shapes and variations; the analysis hints stand down
Endgame	No	Set up one position to study, using the placement keys and Confirm under the board - see section 4

Table 3-1 The three modes

AI side

The two switches combine freely; each of the four combinations has a clear meaning:

AI Black	AI White	Moves first	Notes
On	Off	AI (centre point)	You play White. The AI opens on the centre point and answers your moves
Off	On	You	You play Black. The AI answers after your move
On	On	Both are AI	AI self-play: you just watch, board clicks are ignored; the centre key pauses / resumes
Off	Off	You	Two-player manual play: no AI, you play both colours

Table 3-2 The four AI-side combinations

Thinking time

Item	Value	Notes
Range	0.2 to 300 s in 0.1 s steps	Type the number directly; default 2 s - from lightning-fast to a deep think
Strength	Longer is stronger	The AI candidate heat bands and the scan colours follow the thinking time as well
Synced with Link window	One single value	The Link window's Think Time is this same setting; changing either side updates the other
Relation to background thinking	No overlap	Background thinking only preheats during the opponent's turn and does not eat into this budget
Actual consumption	about 92%	The engine stops at 92% of the budget, leaving room to finish the last iteration; squeezing it harder usually just cuts that iteration in half and is not stronger

Table 3-3 Thinking time: range and effect

Background thinking

The "Background thinking" switch: when on, the engine works out your reply during the opponent's turn (this uses engine-native pondering, the same rhythm as the official bundle), so it is ready the moment you move; when off it stops the moment it moves instead of spinning at high CPU (off by default).

Dynamic time

Dynamic time splits the thinking budget by position difficulty instead of giving every move the same number of seconds. Every signal comes from the engine's own output on the previous move (the score, whether a mate was found, whether the best move keeps changing), so it costs no extra search and adds no latency. The seconds saved go into a time bank that hard positions borrow from; the bank is bounded to −1xN .. +2xN, so once it is used up the schedule falls back to "exactly N per move" and can never snowball.

Position (from the previous move)	Factor	Why
Mate found (+M35 / -M34)	0.40	Attack or defence is clear - more time buys nothing
Big advantage (score >= 2500)	0.70	The result is decided
Clear advantage (>= 900)	0.85	The plan is set, only details remain
Clearly worse (<= -1500)	0.80	Very few defensive choices exist
Level and the best move keeps changing	1.15	Genuinely hard - borrow from the bank
Same best move twice in a row	0.85	The search has converged

Table 3-4 Dynamic time: factors and triggers

Measured with a 2 s setting: level opening moves take 2.2-2.6 s, while positions with a forced win reply in 0.1-0.2 s. Turning the switch off restores exactly N seconds per move and empties the bank.

Rules

Rule	Forbidden	Swap	Key points
Freestyle	None	No	Five or more in a row wins - the most permissive rule
Freestyle, overline loses	None	No	Standard: an overline does not win, exactly five is required
Renju (forbidden moves)	Black	No	Black is forbidden 3-3 / 4-4 / overline; White is unrestricted. Forbidden points are drawn as red crosses
Swap1	None	Yes	After Black plays move 1, White may choose to swap colours
Yamaguchi	Yes	Yes	Black places 3 stones (B1 centre, W2 within 3x3, B3 within 5x5) and declares how many 5th-move picks there will be; White picks a colour and plays move 4 anywhere; Black names the picks (symmetric points may not repeat) and White chooses one as move 5
Taraguchi-10	Yes	Yes	B1 centre, W2 within 3x3, B3 within 5x5, W4 within 7x7; move 5 is either played directly (9x9) or Black calls ten picks with White choosing one, then White picks a colour
Tarannikov	Yes	Yes	B1 centre, W2 within 3x3, B3 within 5x5, W4 within 7x7, B5 within 9x9; after every move the opponent may swap colours. No tentative fifths

Table 3-5 The seven rules: forbidden moves, colour swapping and key points

Lookahead and mate search follow the selected rule. Swap-type rules open a small dialog at the start of the game asking you to choose first or second. Switching rules never clears the board - the position is kept and play continues under the new rule, with the result and the forbidden marks recomputed.

(b) The Engine section



Figure 3-2 Game Setup - the Engine section (threads / hash / engine variant / weights / candidate range / analysis points / max depth / nodes)

Threads The engine search thread count. Steps follow the real topology, with physical/logical core counts, a recommended value and the cap shown beside the drop-down. It starts at the recommendation (performance cores only on hybrid CPUs); raise it toward the cap when cooling and scheduling allow

Hash The engine transposition-table (TT) size. The working engine lane takes 100% of the selected step while the other lanes drop to a floor value, so the total always equals the chosen step

Engine variant "Auto (recommended)" picks the fastest binary from the instruction-set builds by measured throughput

Weights "NNUE (accurate)" for strength, or "classical" for speed in fast games

Candidate range The engine's CAUTION_FACTOR in six steps: 0 most aggressive (5x5 range) ... 5 most cautious (whole board), default 3 (7x7+4, recommended)

Analysis points 1 to 8, default 1 (one full-strength PV); shared with the Analysis card, kept in sync both ways

Max depth / Max nodes Optional hard caps; leave at their defaults for unlimited search

Active / Idle / Usage How the hash is split between the main lane and the helper lanes

Table 3-6 The Engine fields

(c) The Board Display section

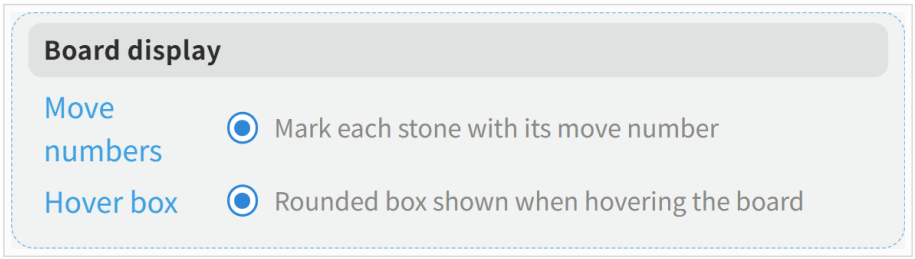


Figure 3-3 Game Setup - the Board Display section (move numbers / preview box)

"Show move numbers" prints the move order on each stone, which is most useful when reviewing. The "preview box" is the rounded square shown under the mouse; it is on by default, and turning it off gives a cleaner board.

(d) The Position section

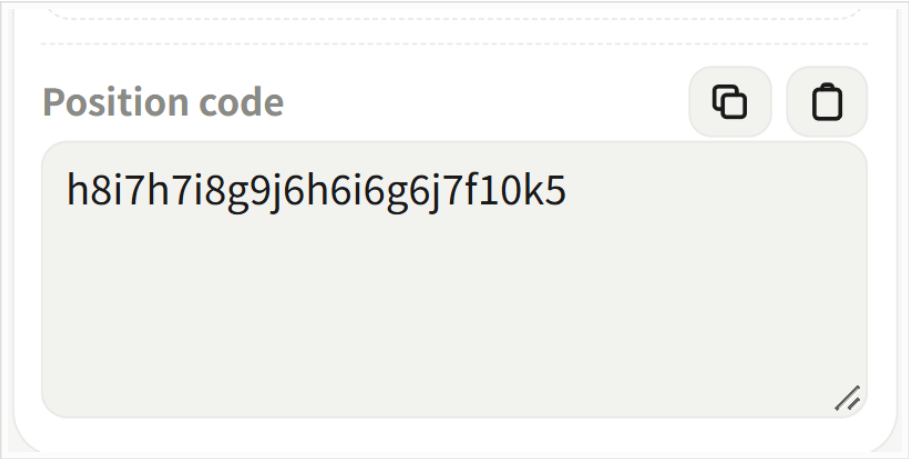


Figure 3-4 Game Setup - the Position section (copy / paste / load a position code)

A position code is a lowercase column letter plus row number, concatenated move by move (the gomocalc.com format). For example **h8i9g7** means H8, then I9, then G7. Use Copy / Paste / Load to import, export, share or archive a position.

(e) The AI-side pop-up



Figure 3-5 The "AI side" pop-up, reachable at any time from the button row under the board

4. Board Operations and Endgame Setup

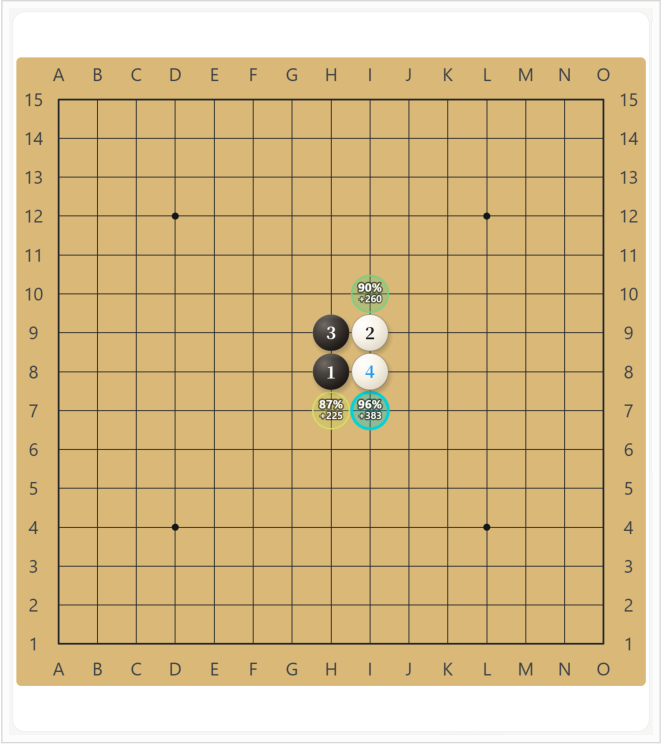


Figure 4-1 The main board: coordinate axes along the top, the button row along the bottom, and one resume / pause key at the centre

Action	How	Notes
Play a move	Click an intersection	The side to move follows the parity of the move count - no manual switching
Previous / next	The ‹ › keys under the board	Step back and forward one move at a time, for review
Restart	The Restart key under the board	Saves the current game to History first, then clears the board
Save position	The Save position key	Stores the current position in History manually
Rotate	The Rotate key	Turns the stones 90 degrees clockwise
Flip	The Flip key	A pop-up offers left-right, top-bottom and the two diagonal flips
Shift	The Shift key	Moves the whole position one cell; if a stone would leave the board nothing moves and you are told
Think one move now / pause	The key at the board centre	When idle, the AI computes and plays one move for the side to move; while thinking, it pauses
AI side / Settings	The two rightmost keys under the board	Open the AI-side and Settings pop-ups

Table 4-1 Board operations at a glance

Rotate, Flip and Shift move the stones only - the grid and the coordinate axes never move.

Endgame setup

Switch the mode to Endgame and placement keys appear under the board, so you can set up an exact position to study:

Key	How to use it
Alternate placement	Black and White take turns, placed in move order
Free placement	Open the small rounded box, pick Black or White, then place that colour freely in any number and any order; the selected key changes colour
Confirm	Press it to fix the position: the key turns pale green and reads confirmed. From then on Restart returns to this position and it becomes the first frame when saved to History; press it again to release

Table 4-2 The three endgame placement keys

In Endgame mode the AI takes no part at all, and swap-type rules are greyed out (an endgame has no opening swap).

5. Calculation and Evaluation

This card is the analysis workbench, in four layers: the four keys - Analyze / Scan / Balance (1-2 points) / Clear; the Routes box, one full route per analysis point; Lookahead, VCF and VCT with the step-through code chain; and Analysis points. Only one of the four calculations runs at a time - switching stops the previous job and wipes its marks, so two sets of annotations never stack up.

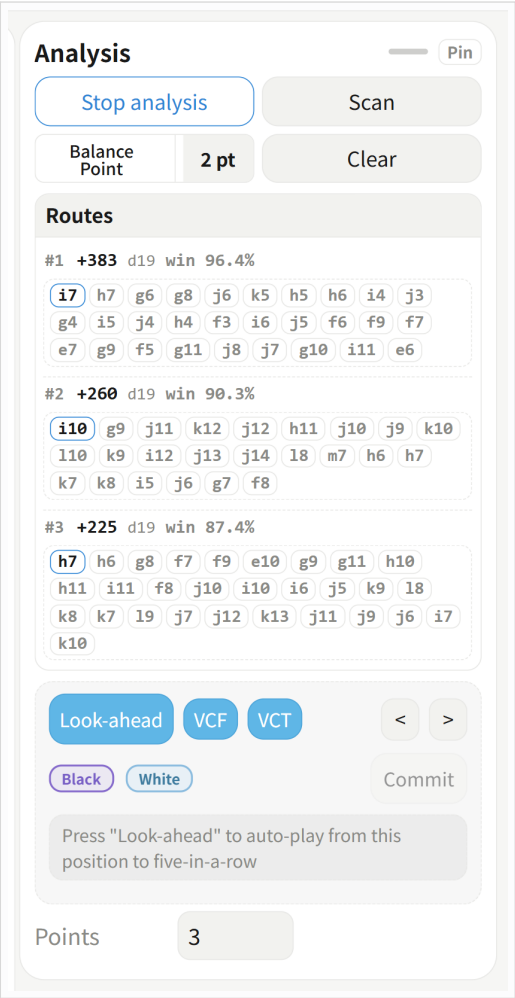


Figure 5-1 The Calculation and Evaluation card: Analyze / Scan / Balance (1-2 points) / Clear, the Routes box, Lookahead, VCF and VCT, and Analysis points

Analyze Keeps deepening the position on the main lane and lists the engine candidates on the board (best point plus up to 7 more), with the Routes box filling in ranked alongside. Press again to stop. Works in Place, Endgame and manual play too

Scan Engine-native YXSEARCHDEFEND over every point inside the candidate range; the percentage in each ring is the opponent win rate if they play there, so higher means defend it. The key then reads Stop and keeps the last frame

Balance Official YXBALANCETWO paired search: candidates with no clear edge for either side, plus the engine score for the pair (|eval| <= 50 is balanced, ScalingFactor 200). Thin solid ring, win rate over score inside, rank badge outside

Points Cycles 1 -> 2 balance points: one is the single swap-2 move, two are the pair. An empty board is fixed at 3 moves

Routes One ranked route per analysis point: a header (rank / score / depth d / win rate) and a full pill-style code line. Hover to preview, click to pin, double-click to play that move

Lookahead, VCF, VCT See the next section; the keys sit mid-card with the code chain and Confirm, and the attacking-side indicator below

Clear Erases every analysis mark this card leaves on the board, leaving the position untouched

Table 5-1 The features of Calculation and Evaluation

Item	Value and meaning
Analysis points	1 to 8, default 1; shared with Game Setup and synced both ways. One PV reaches 27 ply at 10 s on this machine, three reach 25
Balance points	1 or 2, cycled by the key; an empty board is fixed at 3 moves
Scan colours	Rings band by win rate: darker and more solid means more dangerous, and the percentage is printed inside
Routes	A header plus a code line per route; the header carries rank, score, depth (d) and win rate
One job at a time	Analyze, Scan, Balance and Lookahead are mutually exclusive; switching stops the old one and wipes its marks

Table 5-2 Key parameters of Calculation and Evaluation

All four calculations share one engine but use different channels (Analyze takes the long-running main lane, Scan uses YXSEARCHDEFEND, Balance uses YXBALANCETWO), so switching always tears the previous job down cleanly before the next one starts - two channels interrupting each other is what used to make the depth collapse.

6. Lookahead, VCF and VCT

Feature	Description
Lookahead	Plays the position forward move by move until five in a row genuinely appears on the board - it never declares a result from the engine's score, and in the worst case fills the board and calls a draw. The first move uses a reinforced global search to find a breakthrough and an estimated time range is shown; the simulated stones are translucent with the move number on them; every step applies forced-reply checking (make five if you can, block an opponent five, answer a live three)
VCF	Continuous-four forced-win search: only sequences of consecutive fours are considered
VCT	Continuous-four plus live-three threat search: live threes join the threat sequence
Attacking side	The Black / White tags are indicators only (not clickable). The side is decided automatically as the side to move; in mate problems, by convention, the side with more stones
Code chain	Step through it with the arrow keys; hovering previews up to that move; click to select and press Confirm to play that segment onto the board
Ten-call	Asks for 10 candidate points for the opponent to choose from

Table 6-1 The six features of Lookahead, VCF and VCT

VCF and VCT both run a local mate searcher (multi-threaded with iterative deepening). On success the winning line is drawn on the board; if there is none, a dialog states plainly that the attacking side has no such forced win. Once selected, they keep probing while you play and switch into the mate sequence as soon as it can take over.

7. Engine Dashboard and Evaluation Curve

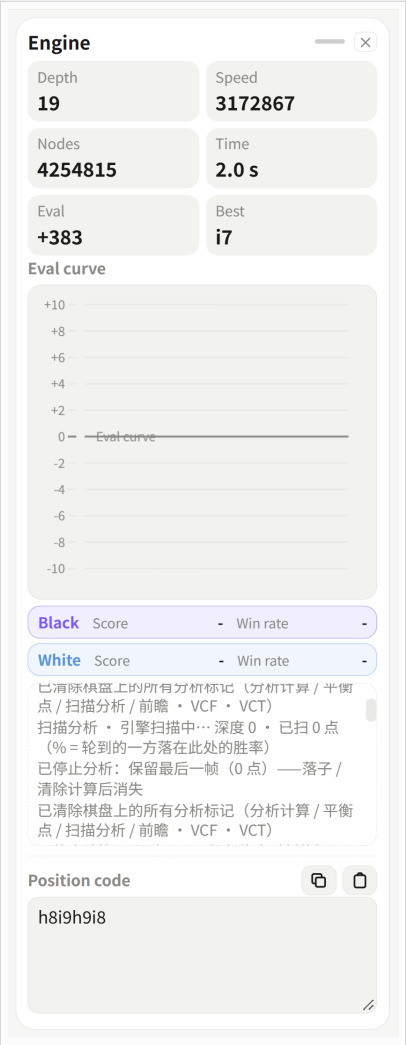


Figure 7-1 The Engine Dashboard and Evaluation Curve card

Depth How deep the current search has reached, in plies

Speed Nodes searched per second - the quickest way to see whether the compute is being used properly

Nodes Total number of nodes expanded in this search

Time Elapsed time (accumulated across turns in Live analysis mode)

Eval The score of the current position for the side to move: positive is better, negative is worse

Best The best move the current search has found

Evaluation curve Records the whole game as a Black line and a White line; the vertical axis scales automatically (up to about 1200) and the horizontal axis scrolls

Status line Your turn / AI thinking / paused / game over / reciting, etc. (at most two lines)

Table 7-1 Dashboard and curve fields

During any calculation (AI move, analyze, scan, balance, lookahead, VCF, VCT) these values update live - the most direct way to tell whether the engine is working and working smoothly.

8. History and Review

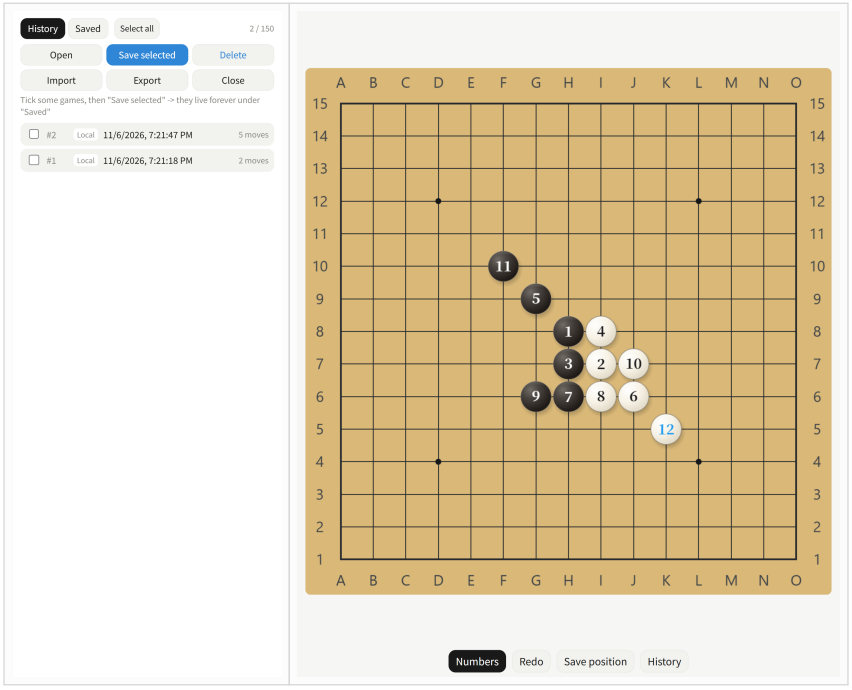


Figure 8-1 The History drawer (left) and the Review window (right): the drawer stores games, the review window is board only

Key / action	What it does
History / Saved	The two pages inside the drawer: automatic archives, and games you saved yourself
Open	Loads the selected game onto the board (VC and Endgame records open with the whole position as the first frame)
Save selected	Moves the selected records into the Saved page
Delete / Select all	Deletes the selected records; Select all ticks everything on the current page
Import / Export	txt import and export are supported
Ctrl / Shift multi-select	Continuous or scattered multi-selection, like a file manager
Right-click menu	On a single record: open the review, rename, export that game, store it in Saved History or delete it
Review window	The top-bar Review button opens a standalone board-only window that never talks to the engine; Recite / Replay review steps through the game move by move

Table 8-1 History drawer and Review window operations

9. Screen Recognition (Gomoku Vision)

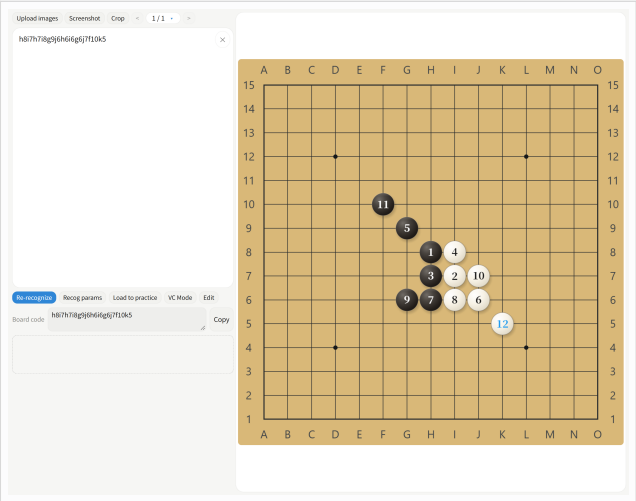


Figure 9-1 The recognition panel: upload / screenshot, framing, recognition, manual editing and the image drawer

Upload / screenshot	Import a screenshot of a game, or capture a screen region directly
Frame	The selection box snaps to the board and leaves about 1/30 of the board as margin so the outermost stones stay complete; you can also frame the image region by hand before recognising
Recognize	Press Recognize to start; the result becomes a position code that can be copied or loaded for practice
Edit	Manual fixes: swap Black and White (click one of each), delete stones, add Black or White stones; Ctrl+Z undoes
Recog params	The "Recog params" key lays the five thresholds (grey offset / black offset / black percent / white offset / white percent) directly on the panel and applies them while you drag; the hint line has a fixed height, so the layout never jitters while you move a slider
Image drawer	Every uploaded image goes into the drawer automatically (the oldest are dropped past 150), with multi-select, delete and copy

Table 9-1 How to use screen recognition

Recognition is specifically hardened for the "it looks recognisable but will not recognise" cases. When the whole board reads empty the thresholds are first relaxed against this frame's own peak; if it is still empty a local-contrast reader takes over (it only compares each intersection's centre patch with the diagonal patches around it, so dark and washed-out boards no longer drop stones). Board finding tries four lanes in turn: the original image, JPEG de-blocking, CLAHE equalisation and a 1.5x upscale. If the grid does not line up with the lines the older projector-based recogniser (far more tolerant of rotation) re-checks and the more plausible result wins; a whole row or column of 15 stones is treated as a one-cell misalignment and re-read after shifting the grid.

Item	Normal recognition	VC (mate problem) mode
Black / White counts	Must be equal	May differ
Add / delete stones	Constrained by the game	Add Black or White without limit; delete freely
After loading	Play continues from the position	Switches to Endgame mode and presets the attacking side (the side with more stones)
Mate search	Not started automatically	Also not automatic - press VCF or VCT to decide

Table 9-2 VC mode versus normal recognition

10. Rules, Language and Appearance

Item	Notes
Rules	The seven rules listed in section 3 (Game Setup / Rules)
Language	The top-bar EN / 中 button switches the whole UI in one click, including the screenshot-box buttons and the date format in History
Colour	The palette window manages the board colour, the page colour, the dark and light themes and the preset schemes
Layout	The right-hand cards can be dragged by the title bar to reorder, folded with the small dash and hidden with the close key; the layout is remembered for next time
Stone material	Settings -> Font -> stones: Realistic (gradient plus a soft contact shadow, default), Jade (layered pale tints with a glass sheen) and Flat (fast single colour)
Board sharpness	The board renders at 2x supersampling, so stones, move counters, grid lines and coordinates stay crisp on high-DPI screens; shadows are baked sprites, about 0.4-0.6 ms per repaint

Table 10-1 Rules, language and appearance

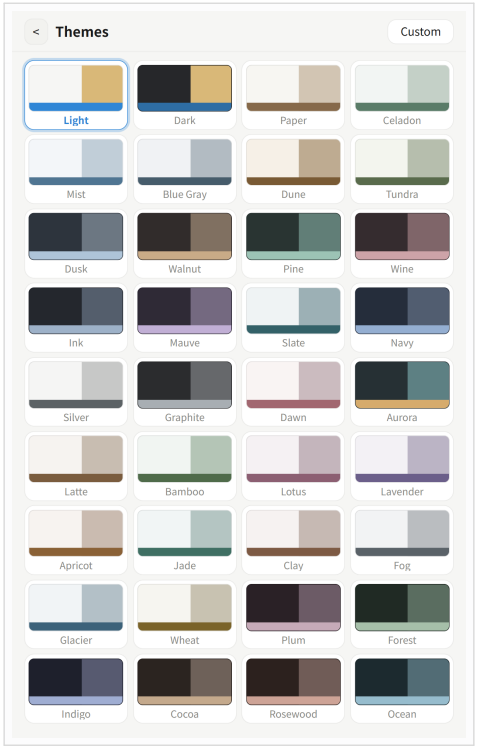


Figure 10-1 The palette window: board and page colours, themes and presets

11. Link Mode (assistance inside another playing app)

Link mode is the heart of this release. It puts both "look at the board" and "think about the board" into one fixed-size small window: it reads the game you are playing through screen recognition, then lets the engine think for you. It touches none of the other app's internals - it does exactly two things: capture the screen and simulate clicks.

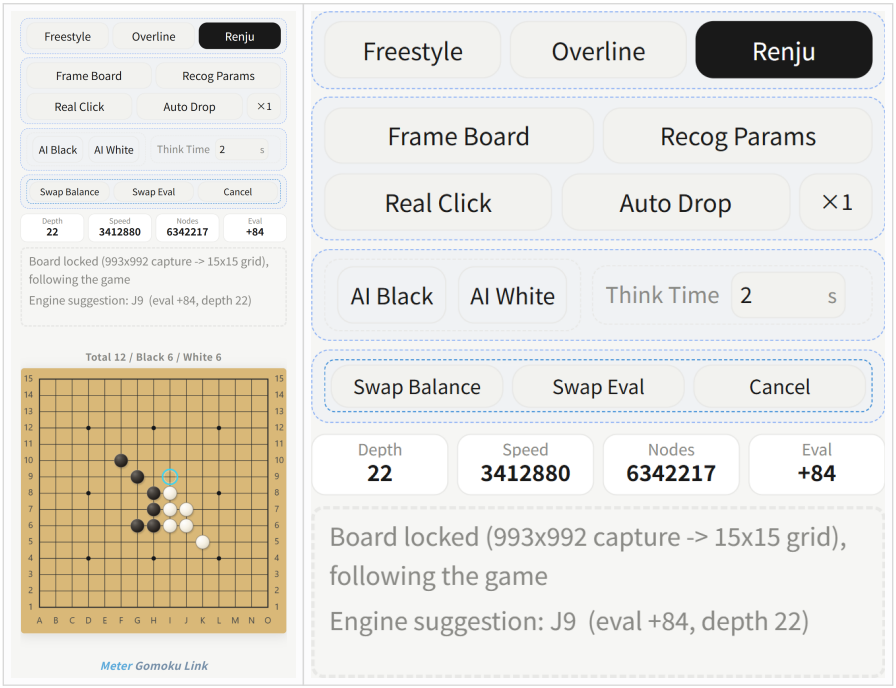


Figure 11-1 The Link window (left: whole window; right: the control panel enlarged) - the bar, the panel, the status area and the position mini board on the right

(a) Framing and recognition

Key	What it does
Frame Board	Drag a rectangle over the target board on screen; recognition starts once the frame is set
Recog Params	Replaces the button area above with the five recognition thresholds (grey offset / black offset / black percent / white offset / white percent); dragging applies them live, Apply saves and folds the panel, Cancel restores the values that were last applied
Real Click	Clicks with the real mouse cursor (off by default). Only needed for apps that accept nothing but real mouse input (QQ Gomoku, for example); while it is on it takes over your cursor
Auto Drop	When on, the app thinks as soon as it sees the opponent move and plays for you through simulated clicks (see below)
x1	Clicks per move: 1 to 3, cycling (the tooltip shows the current step)

Table 11-1 Framing and recognition keys

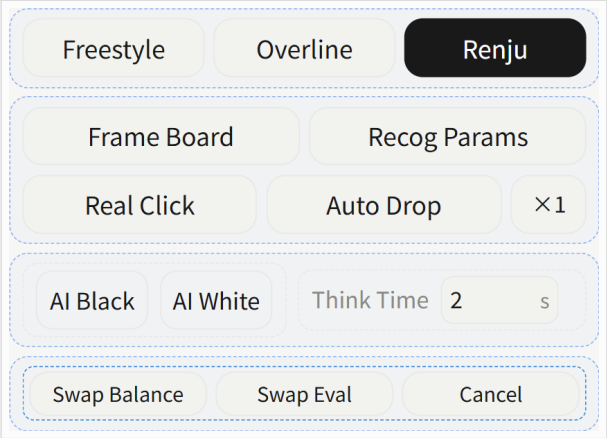


Figure 11-2 The Link window bar: Frame Board / Recog Params / Real Click / Auto Drop / x1

Recognition runs quietly in the background in two steps. The first frame is pinned down by the app's own detector, which also reports the exact position of all 15 horizontal and 15 vertical lines; once locked, the recognition engine's synchronous stone reader takes over, and the window fingerprint is handed across at the moment of the hand-over - so the full search happens once and the very next frame already continues from it. From then on it works cell by cell: every intersection keeps a 6x6 sample signature, so a still picture re-reads 0 cells at almost no cost, and a single stone re-reads exactly one cell. That is why the frame colour never flashes back and forth and why recognition costs no CPU at rest. When the crop box is auto-tightened it leaves 0.8 cells of margin outside the board - the outermost stone on a square board is exactly the one that used to get clipped, and this margin keeps it complete without pulling in neighbouring UI.

Three fallbacks make it tolerant of awkward pictures, and they run only while every threshold is still at its default - any non-default value switches recognition to the strict path. If the whole board comes back empty, the thresholds are relaxed once against this frame's own peak; if it is still empty, a third local-contrast reader takes over (it only compares the brightness of each intersection's centre patch with the diagonal patches around it, so dark

boards and washed-out boards are treated the same); and if the grid does not line up withthe lines (low liveness) the older projector-based recogniser, which tolerates rotation muchbetter, re-runs and the more plausible result wins. A whole row or column of 15 stones -impossible in Gomoku - is treated as a one-cell misalignment and re-read after shifting thegrid by one.

Recognition parameters

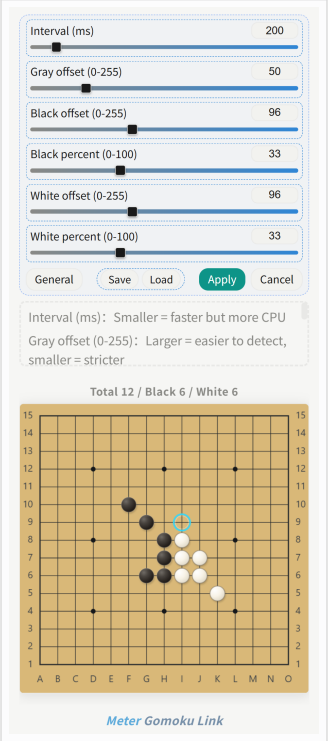


Figure 11-3 Recog Params: one click swaps the button area for the five thresholds, applied while you drag

Parameter	Notes
Grey offset (0-255)	The grey threshold for "there is a stone here". Higher detects more stones, lower is stricter
Black offset / percent	Grey threshold and in-cell coverage for Black. Too many Black stones -> lower; missing ones -> raise
White offset / percent	The same two values for White
Strict parameters	If any value differs from its default, recognition switches to the strict path: the result follows exactly what you set and is no longer overridden by the relaxed / local-contrast / grid-check fallbacks; the fallbacks run only while every value is at its default
General / Save / Load	General loads the defaults (50 / 96 / 33 / 96 / 33); Save and Load store the current five values under a theme name (up to 50) and the drawer offers Load / All / Delete / Undo
Apply / Cancel	Apply saves and folds the panel; Cancel restores the five values to what they were when the panel was opened, including the values in effect in the host

Table 11-2 The five thresholds and four keys of Recog Params

(b) The control panel

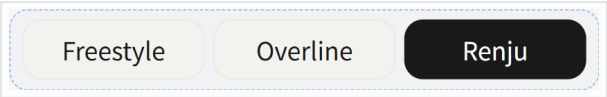


Figure 11-4 The three rule keys: Freestyle / Overline / Renju

Control	Notes
Rule keys	Freestyle / Overline / Renju - switches the endgame and forbidden-move rules used by the Link window
Frame Board / Recog Params / Real Click / Auto Drop / x1	See the previous section
AI Black / AI White	Mutually exclusive (you may also select neither and play yourself). The choice is mirrored unconditionally into Game Setup in the main window
Think time	0.2 to 300 s, the same single value as the main window - the two are always in step
Swap Balance	Gives opening balance points (see below)
Swap Eval	A feature independent of Swap Balance that scores the current swap position to help you decide whether swapping is worthwhile
Cancel	Stops an in-flight Swap Balance / Swap Eval and clears the balance markers already drawn, returning the window to its normal state

Table 11-3 The Link control panel

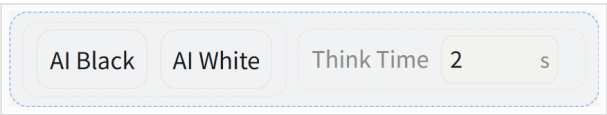


Figure 11-5 AI Black / AI White and Think time (mutually exclusive; the choice mirrors to the main window)

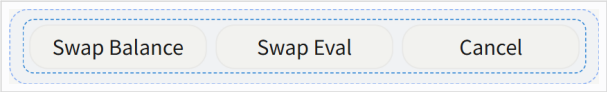


Figure 11-6 Swap Balance / Swap Eval / Cancel (Cancel is only active while a calculation or markers exist)

(c) Status area and position mini board



Figure 11-7 Status figures: Depth / Speed / Nodes / Eval



Figure 11-8 The status text area (a dashed box with room for three lines or more, so long English wraps never fold up)

Area	Contents
Status figures	Depth / Speed / Nodes / Eval, updating live during any calculation
Status text	Says what is happening (recognising / thinking / balance points marked, ...). While Recog Params is open it shows the six parameter hints instead, scrollable
Position mini board	Mirrors the recognised position live, with the move count beside it. It follows the same rule as the assistant position window: it shows what the current frame recognised and keeps the last good position when nothing is recognised - it is never cleared for no reason

Table 11-4 Status area and position mini board

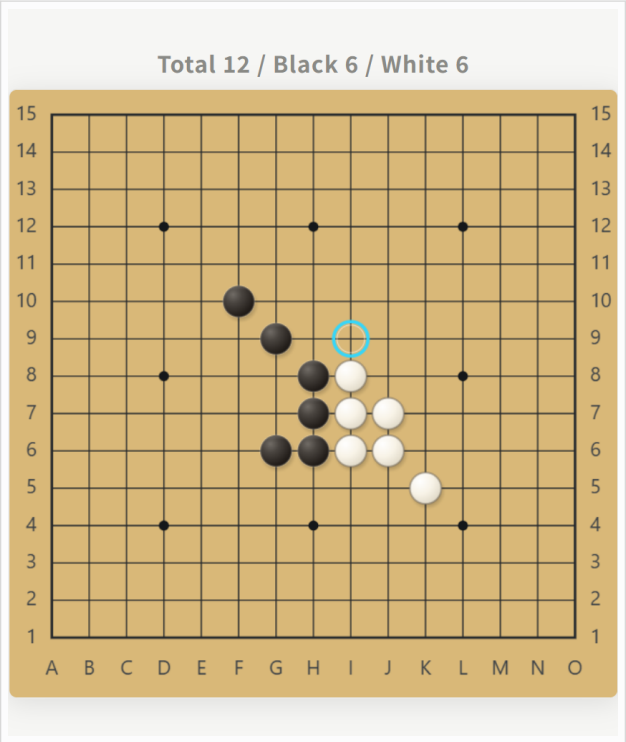


Figure 11-9 The position mini board (mirroring the recognised position) and the move count

(d) Division of labour

The Link window owns all recognition and all AI work (evaluation and auto drop). The main board only mirrors the recognised position and the AI-side setting and never evaluates or drops, so the two can never fight over the same compute. Opening Link mode switches the main window to a 1.16 : 1 layout; closing it restores your own mode and settings.

(e) Auto Drop

Stage	Method	Trigger
1	PostMessage straight to the target window	The default first choice: lightest, and it never disturbs the physical mouse
2	SendInput	Escalated automatically when the next frame says stage 1 failed
3	SetCursorPos + mouse_event (a real click)	The last resort when both earlier stages fail; once it works it is pinned and reused
Verdict	Next-frame confirmation	Every stage is judged by the following frame; only a failure escalates
Retry	A whole round restarts (up to 3 rounds)	All three stages failing no longer means waiting forever: a full round restarts automatically, and only three failed rounds stop and tell you
Finish	Ends with the game	A finished game (five in a row) switches Auto Drop off; the next game needs it turned on again

Table 11-5 The three-stage click pipeline and its retry

If you play Black and the board is empty when Auto Drop is switched on, it plays the centre opening move for you. Your physical mouse is never disturbed - only the last stage borrows the cursor, and it puts it back immediately.

(f) Swap Balance

This is for swap openings: the app computes a set of opening points that are as fair as possible for both sides and draws them on the matching intersections inside the recognition frame. On an empty board it uses the official opening book: the engine first returns a three-stone opening for the current rule (the Swap2 book for Freestyle, the RIF book for Renju; if the engine is unavailable the app falls back to its own matching tables - 9 Freestyle sets and 26 Renju sets, balanced lines only, no winning lines), which is then expanded through the board's eight symmetries (identity, main diagonal, left-right, top-bottom, 180 degrees, clockwise and anticlockwise 90 degrees, anti-diagonal) and de-duplicated, so many more shapes are reachable than the handful listed in the table. Once the three stones are laid out the balance engine scores that pair and reports it - that number is the checkable |eval| (measured 1 to 17 for the RIF sets and 5 to 19 for the Swap2 sets, all inside the official balanceWindow of 50).

Stones on board	Points given	Colours	Notes
0 (empty board)	3 points	2 Black, 1 White	The first stage of a swap opening: Black 1 is fixed at the centre and White 2 / Black 3 come from the official book
3 stones (5 moves total)	2 points	1 Black, 1 White	The second exchange: one paired balance search returns the Black and the White point together (a paired search is the official notion of " balanced"; a single-point search drifts to -350~-470 on a near-empty board)
More than 5 stones	Not used	-	Past the swap window - the feature reports that it is unavailable

Table 11-6 Swap Balance behaviour by move count

Interaction	Notes
Exclusive with AI side	Pressing Swap Balance automatically cancels any selected AI Black / AI White, because they would interfere
Interaction with Auto Drop	Press Swap Balance first, then turn Auto Drop on: if Auto Drop is already on when the evaluation finishes the points are played automatically in order; if Auto Drop is turned on only after the evaluation, the points are played in order the moment it is switched on
The Cancel key	No patience left halfway through? It stops the balance calculation in flight, clears the balance markers and the queued dropping, and returns to the normal Link state (it will not finish drawing the marks afterwards)
Marker lifetime	The markers stay inside the recognition frame; the balance engine is released immediately and stops using compute; once you have played those stones the markers disappear automatically

Table 11-7 How Swap Balance interacts with the rest

(g) On-board marks

Mark	Appearance	Meaning
Suggested move	Sky-blue ring	The move the engine recommends
Swap Balance - Black	Purple ring + number	A Black point of the 2-Black-1-White set; the number 1 / 2 / 3 is the order of play
Swap Balance - White	Blue ring + number	A White point of the same set
Capture protect ion	Excluded from screenshots	Every mark is drawn directly on the matching intersection inside the recognition frame and is excluded from screen capture - it can never be misread as a stone by the app's own recognition

Table 11-8 The marks drawn on the board

(h) Performance: how the CPU is shared

In link mode recognition and the engine run at the same time, so the app puts the two workloads on different physical cores. That is why a long session neither stutters nor loses strength.

Subsystem	Share	Notes
Engine search	About 2/3 of the physical cores	Split by physical core rather than logical thread, so hyper-threading never causes contention. The exact thread count appears under "threads / cap" in the Engine section
Screen recognition	The remaining 1/3	Recognition only reads when something changed: almost no CPU while the picture is still, and one stone only re-reads that one cell
Engine hash	Equal to the chosen step	The working main lane takes 100% of the transposition table while the other lanes drop to a floor value, so the total always equals the step you picked in Game Setup; the cap is derived from real memory
Long search stability	Converges automatically	During a very long search the marks and the evaluation curve keep only the most recent fixed number of entries, so memory use stays flat

Table 11-9 How engine and recognition share the CPU

(i) The window stays on top

The Link window itself is pinned to the top of the desktop: it is created as a topmost window, pushed back to the top whenever its position changes, and re-raised when it is shown or when the display state changes - so the app you are playing in cannot cover it. The app does not interfere if you minimise it yourself; it only re-raises when you bring it back.

Finishing up	Notes
Closing the Link window	The main window restores your own mode and settings, and play, analysis and everything else keep working as before
Exiting the app	Every engine and related process is shut down at the same time, leaving nothing behind

Table 11-10 Closing the Link window and exiting

12. Frequently Asked Questions

Symptom	Cause	What to do
Board not recognised, or recognition unstable	The frame does not fully enclose the four corners and the border	Re-run Frame Board and leave a little more margin; on a crowded board the app combines the stones and the border to decide
Stones on the board edge are missed	Not enough crop margin, so the outermost stone is clipped	The automatic crop already leaves a 0.8-cell margin, which targets exactly that outermost stone on a square board. If it still happens, include the board frame itself in your selection
Auto Drop is on but no move is ever played	One failed three-stage click used to stop the cycle for good	After all three stages (PostMessage, then SendInput, then a real click) fail it starts a fresh round automatically, up to 3 rounds, and only then stops and tells you. If it still does not move, press Frame Board to re-lock the board
Strength varies, sometimes weak	The engine and recognition competing for the same CPU and memory and occasionally starving each other	The app splits the physical cores 2/3 to the engine and 1/3 to recognition, and derives the hash table from real memory. Raising the thinking time also helps immediately
An occasional "engine response timeout"	Rare, only in the instant when the engine is pre-empted or switching gears	The app re-evaluates automatically about 1.5 s later and retries silently; normal play is unaffected
Stuck on "recognizing"	A hollow selection window, or the first-run warm-up	Close the hollow window and re-frame the board; recognition warms up on first open, so wait a moment
Nothing happens when Swap Balance is pressed	The current move count is not inside the swap window	Swap Balance is available only on an empty board (3 points, 2 Black 1 White) or with 3 stones present (2 points, 1 Black 1 White). Past five stones the app says plainly that it is outside the swap window
Is the Internet required?	-	No: the app and the engines run entirely on your machine and listen on no network port
Leftover processes after closing?	-	When the app exits it shuts down every engine and related process, leaving nothing behind

Table 12-1 Frequently asked questions and answers

13. Open-Source Components and Licences

The software is built from several components, each distributed under its original licence. The tables below list every third-party / open-source component shipped with the software, together with its licence and official link, so that everything can be verified and the source obtained.

Component	Purpose	Licence	Official link
Meter Gomoku interface and companion code	Interface, host program and companion scripts (author: Meter)	MIT	Licence notice included with this manual
Rapfi	Playing core (Gomoku / Renju engine); shipped in five instruction-set variants with weights	GNU GPL-3.0	github.com/dhbloo/rapfi
OpenCV	Board and stone recognition, statically linked into GomokuVision.exe with no extra DLL	Apache-2.0	opencv.org
Microsoft Edge WebView2 Runtime	Desktop UI runtime, provided by Windows / Edge and not redistributed by this app	Proprietary (Microsoft)	microsoft-edge/webview2
Source Han Serif / Sans (Noto Serif SC / Noto Sans SC)	Typesetting fonts for this manual, embedded in the document	SIL OFL 1.1	adobe-fonts
Inno Setup	The tool that builds the Windows installer (used for packaging only, not distributed)	Inno Setup License (free for commercial use)	jrsoftware.org/isinfo.php

Table 13-1 Third-party / open-source components shipped with the software

Licence	Permitted / required / link
MIT	Permits obtaining, using, copying, modifying, merging, publishing, distributing, sublicensing and selling copies free of charge. Requires the original copyright and licence notices to be kept in the copies. The software is provided "as is", without warranty of any kind, express or implied
GNU GPL-3.0	Permits obtaining, using, modifying and redistributing free of charge. Requires the licence and a way to obtain the source to be kept when redistributing (Rapfi source: github.com/dhbloo/rapfi; full text: gnu.org/licenses/gpl-3.0)
Apache-2.0	Permits commercial use and redistribution. Requires the copyright and licence notices to be kept and modifications to be stated. Full text: apache.org/licenses/LICENSE-2.0
SIL OFL 1.1	Permits free personal and commercial use, including embedding into redistributed documents. Requires that the font files themselves are not sold on their own and that modified versions are not sold under the original font names. Full text: openfontlicense.org
Proprietary (WebView2)	Used under Microsoft's own terms; this installer does not redistribute it and only offers the official download page when it is missing

Table 13-2 What each licence permits, requires, and where to read it

Compliance notes

(1) Rapfi - this app calls Rapfi as a compiled separate process; the two communicate only through standard input/output and messages and are not merged into a single work. When Rapfi is distributed with this app, its GPL-3.0 licence text and a source notice are kept alongside it (in the rapfi-native/ folder).

(2) OpenCV - statically linked into GomokuVision.exe; no OpenCV dynamic library is distributed separately. Its Apache-2.0 licence requires the copyright and licence notices to be kept, and they are kept with the recognition component.

(3) Fonts - neither this manual nor the software uses any non-commercial font (such as Microsoft YaHei or SimHei, both Windows system fonts). Everything is typeset in the open-source, same-origin Source Han Serif / Source Han Sans.

(4) Development-only dependencies - the scripts used to build and package the app and their dependencies (NumPy, Pillow, PyMuPDF, reportlab, ...) run on the build machine only and are not distributed with the software, so they are not listed above.

(5) Disclaimer - the software is provided "as is"; the author accepts no liability for its fitness, its stability, or any direct or indirect loss. Please use the assistance features in a way that respects the rules of the platform you play on. All component names and trademarks belong to their respective owners.

Author: Meter. The interface and companion code are released under the MIT licence; the playing core Rapfi under GPL-3.0; the recognition component and OpenCV under Apache-2.0; the manuals are typeset in Source Han Serif / Source Han Sans (SIL OFL 1.1). Download: https://wwbbj.lanzout.com/b016l49aqj Password: 15